

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

INTRODUCTION

The Philippines is one of the world's leading banana-producing countries, supported by a tropical climate that favors the cultivation of major varieties such as Cavendish, Saba, and Lakatan. Banana is a cornerstone of the national agricultural economy, providing livelihoods for millions of smallholder farmers and contributing significantly to export revenues. Beyond its economic value, banana is a staple food consumed worldwide and is valued for its potassium, dietary fiber, and vitamin content, which support heart health and sustained energy. However, the farmers who depend on banana cultivation are increasingly threatened by diseases that can devastate their crops. Among these, Cavendish was threatened by Fusarium wilt (FW) tropical race 4, a disease caused by a soil-borne pathogen, *Fusarium oxysporum f. sp. cubense* (Foc), which colonizes the plant's vascular tissue through the roots (ProMusa, 2025). Banana Bunchy Top Virus (BBTV), meanwhile, is considered the most destructive viral disease of bananas in the country. It severely affects the Lakatan variety, which is commonly grown by small-scale growers. The industry further faced a lack of supply, particularly in the banana chips industry, mainly due to traditional Saba taking several months (18-24 months) before it can be harvested. (Mcbautista, 2021)

Local Data from the Office of Agriculture in Rosario, Batangas indicate that the most prevalent threats are Sigatoka, BBTV, Panama Disease (Fusarium Wilt),

and Bacterial Wilt. These diseases persist year-round, with heightened transmission during the rainy season or in areas with high aphid populations. If not identified early, they can spread quickly, reduce photosynthesis, and lower fruit quality, leading to yield loss of up to 50%. Among these threats, this study targets three fungal leaf diseases whose symptoms are visible on the leaf surface and can therefore be detected through image-based analysis: Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight. These three diseases have also been studied together in published image based work (Arman et al, 2023; Bhuiyan et al,2023).

Many farmers remain unaware of early infection signs and still rely on subjective visual observation or delayed consultation with agricultural experts. Because banana diseases often first appear on the leaves, early detection at the leaf level is critical. Machine learning techniques have been widely applied to banana crop monitoring and disease identification (Sahu et al.,2022)

Black Sigatoka, also known as black leaf streak, is the most economically important leaf spot diseases of banana, affecting a wide range of cultivars than the other Sigatoka Diseases (Esguera et al.,2024). It often defoliates vulnerable cultivars before the fruit branches mature. It is caused by the airborne fungus *Mycosphaerella fijiensis* Morelet (*Pseudocercospora fijiensis*), which spreads from plant to plant through wind, rain, and irrigation water splashes. The fungus impedes photosynthesis by causing blackened lesions on the leaves and can eventually kill the entire leaf. The necrotic areas dry out turns brown, then become much darker when soaked by rain, giving the field the appearance of dead black leaves. The leaf edges, which are often most affected parts, tend to fold on themselves. High rainfall

and humidity favor disease development (Fajardo et al.,2022). Without control Black Sigatoka can reduce fruit yield by an estimated 30 to 50%, depending on climate and disease severity. Controlling it requires intensive fungicide programs, with up to 55 applications per year reported in the Philippines (Business Queensland, 2026)

Another leaf disease that will be discussed is Cordana leaf spot (CLS), also known as banana diamond leaf spot, is caused by fungi of the genus *Cordana*: *C.musae* (*Necordana musae*), which is more common in tropical regions, and *C.johnstoni*, which is found in the cooler banana-growing regions of Australia. Although widespread, CLS is not considered a major pest of bananas and is usually a secondary invader of lesions cause by other fungi. Frequent rain and hot humid conditions accelerate its growth and spread. Its spots, which can reach 10 cm, are oval to diamond-shaped and pale brown with a yellow border, and appear on the upper leaf surface. Plants are more susceptible under adverse conditions such as water logging, drought, nutritional deficiencies, or injury or infection from other diseases, which provide an entry point for the fungus. Cavendish bananas appear to be more tolerant to CLS than native banana types such as plantain (Crop Science Australia, 2025)

Pestalotiopsis leaf light is caused by Pestalotiopsis species, which are necrotrophic pathogens capable to infecting a wide range of host plants. Studies show that fungi in this genus produce enzymes and toxins that degrade plant cell walls, aid nutrient acquisition, and induce tissue necrosis. They are responsible for many diseases, including leaf spot, leaf blight, canker, trunk disease, rot, dieback, and various post-harvest diseases. Morphological, Pestalotiopsis species produce

fusiform conidia with four septa (five cells), in which the basal and terminal cells are hyaline and the median cells are brown. (Yu et al., 2025).

To address this problem, this study proposes an Android application for banana leaf diseases detection that allows farmers to simply take a photo of a banana leaf and instantly receive a diagnosis, including the specific diseases detected and recommended management actions, without needing internet access or the presence of an expert. At the core of the study is a comparative evaluation of two machine learning classifiers, Support Vector Machine (SVM) and Random Forest. Both algorithms are well-established in plant disease classification and are lightweight enough to run on a standard Android smartphone, making them practical for deployment in remote agricultural areas. Before classification, leaf images pass through a structured processing pipeline. HSV color space conversion capture color-based disease indicators, Gray Level Co-occurrence Matrix (GLCM) analysis extracts texture patterns from the leaf surface and K-means clustering segment diseased regions from healthy tissue, so that the classifier focuses on the most relevant areas of the image. This study puts accurate, expert-level diagnosis directly into the hands of Filipino smallholder farmers, empowering them to detect diseases early, respond quickly, and protect both their crops and their livelihoods.

STATEMENT OF THE PROBLEM

This research aims to develop an early detection system for banana leaf diseases using image processing and machine learning by addressing the following questions:

1. Which machine learning classifier, Support Vector Machine (SVM) or Random Forest performs better in detecting and classifying banana leaf diseases from digital images, and how can the superior algorithm be integrated into a complete image processing framework?
2. How can OpenCV's feature extraction be combined with the selected classifier in Android environment to enable reliable, offline image analysis and disease localization using a smartphone's hardware?
3. How effectively can the integrated application translate diagnostic results into localized treatment recommendations, assisting farmers in responding to banana leaf diseases promptly?

OBJECTIVES OF THE STUDY

The main objectives for developing this system are:

1. To evaluate and compare the classification performance of Support Vector Machine (SVM) and Random Forest on banana leaf diseases images to identify the superior model.

2. To develop and build an Android application using OpenCV and the better-performing classifier that operates fully offline using smartphone camera input.
3. To evaluate and assess how well the application provides disease diagnosis and actionable treatment recommendations for farmers.

SIGNIFICANCE OF THE STUDY

The researchers believe that this study will be beneficial to the following:

For Farmers: This study provides a practical and affordable tool for early diseases detection, specifically designed for those in rural areas with limited internet access. By using the “one-click” camera feature, farmers can receive instant diagnoses and treatment advice, allowing them to take immediate action. This reduces crop mortality, lowers the cost of unnecessary chemical use, and protects their primary source of income.

For Agricultural Experts and Researchers: This study serves as a fast and efficient supplementary tool for field monitoring. It allows experts to focus on more complex cases while using the system’s data as a foundation for further research in plant pathology. It also demonstrates how traditional machine learning can be optimized for real-world agricultural challenges.

For Government and Agricultural Agencies: This study supports government and agricultural agencies not through large-scale digital monitoring, but by directly empowering the farmers those agencies serve. When individual farmers in remote

areas can independently identify leaf diseases early and apply the correct treatment, the burden on agricultural technicians and extension officers is reduced. Agricultural agencies can reference this study as evidence that lightweight, offline tools are viable for grassroots disease management, and use its findings to inform community-level training programs or future policy recommendations on mobile-based agricultural technology.

For Computer Science Students and Developers: This study serves as a valuable reference for those interested in Mobile Development. It provides a clear example of how image processing (HSV, K-means) and machine learning (SVM, Random Forest) can be integrated into a functional Android Application to solve a specific, real-world problem.

For the Field of Precision Agriculture: This study contributes to the development of inclusive AgTech. It proves that advance computational techniques can be made accessible and lightweight for mobile platforms, encouraging the shift toward data-driven farming practices that are practical for smallholder growers.

For the Institution: This research highlights the commitment of the Bachelor of Computer Science in Computer Science program to innovation and community service. By addressing a local agricultural crisis through technology, it reinforces the institution's reputation for producing socially responsible and technically skilled graduates.

For Future Researchers: This study provides a foundational framework for applying offline machine learning in agricultural contexts. It offers important

methodologies regarding feature extraction (GLCM and Color Moments) and model optimization that can guide similar research in detecting diseases in other crop varieties.

SCOPE, AND LIMITATIONS OF THE STUDY

This study covers the development of an Android mobile application designed for the early detection of banana leaf diseases using image processing and machine learning. The researchers choose to build this for Android platform because Android smartphones are affordable, and widely used, making the application a practical tool that farmers can easily carry and use in the field. The application serves as a diagnostic tool and to identifies a disease at the early stage of visible symptoms on the leaf surface, before the infection spreads and causes extensive damage to banana plant. Farmers can capture photo of a banana leaf using their smartphone camera and instantly receive a diagnosis along with treatment recommendations. The application is designed to work offline, so all processing is done on the user's device without an internet connection. The study also includes a comparative evaluation of Support Vector Machine and Random Forest to determine which classifier performs better for this purpose.

The limitation of the study is that the application can only detect diseases that show visible symptoms on the leaf surface and it does not measure disease severity or infection stage. Detection is limited to three banana leaf disease, namely Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight, which are commonly

found in the Philippines. Other diseases outside this scope, will not be recognized by the system. These include soil-borne and vascular diseases, as well as diseases affecting the roots, stems or fruits, and internal infections. If a plant has both a leaf disease and another disease, the application reports only the leaf-surface symptoms visible in the captured image and does not provide a complete disease profile of the plant. The results may therefore not represent the views of all banana farmers in the province. The model is trained on a fixed datasets of healthy and infected banana leaf images and may not perform equally well on disease variations or banana varieties not represented in the training data.

CHAPTER II

REVIEW OF RELATED LITERATURE AND RELATED STUDIES

This chapter offers an in-depth review of previous research and literature in order to identify gaps, strengths, and limits in the topic of Development of an Android Application for Early Detection of Banana Leaf Disease using Image Processing and Machine Learning. This chapter aims to conduct an extensive literature review with the aim of identifying current research gaps and potential topics for additional study.

THEORETICAL BACKGROUND

The purpose of this study is to develop an early detection system for banana leaf diseases using image processing and machine learning. This system aims to

identify infected leaves by analyzing visual symptoms such as discoloration, spots, and texture variations while providing an efficient diagnosis to support farmers in disease management. By leveraging traditional machine learning techniques, specifically Support Vector Machine (SVM) and Random Forest classifier, the system provides a lightweight and efficient solution suitable for mobile deployment without requiring internet connectivity.

Traditional identification methods involve manual inspections, which is time-consuming, subjective, and often ineffective for large-scale farming. Therefore, the integration of automated image processing and machine learning-based detection systems is essential for improving early diagnosis and control measures and control measures without the need for constant internet connectivity.

The proposed approach involves multiple steps using image processing techniques and machine learning models. These steps include (1) Image preprocessing for resizing, noise removal, and color space conversion using OpenCV; (2) Segmentation of healthy and infected leaf regions using K-means Clustering and Otsu's Thresholding; (3) Feature extraction where diagnostic characteristics are pulled via Color Moments and Gray-Level Co-occurrence Matrix (GLCM); and (4) Classification using SVM and Random Forest to determine the specific disease.

Computer vision, a branch of artificial intelligence, enables machines to interpret and process visual data from digital images. In this study, OpenCV acts as the core engine that allows the smartphone to “see” the leaf. By using Image Processing to clean the visual data and SVM or RF algorithm to categorize it, the

system replicates the diagnostic capability of an agricultural expert. Unlike deep learning models, that requires high-end servers, the combination of handcrafted feature extraction and SVM or RF allows for a lightweight, offline solution.

For offline analysis, the system processes input images captured via a smartphone camera. The model provides diagnosis and management tips, ensuring that farmers can take immediate action. The use of SVM or RF classifier and image processing techniques enable the system to achieve high precision and robustness in detecting the three target leaf diseases: Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight. Diseases such as Panama Wilt (*Fusarium* Wilt), Banana Bunchy Top Virus (BBTV), and Bacterial Wilt are not within the detection scope of this system, as these diseases primarily manifest through vascular, systemic, or viral symptoms that are not reliably diagnosed through leaf surface imagery alone.

REVIEW OF RELATED LITERATURE

OPENCV (OPEN-SOURCE COMPUTER VISION)

Computer vision in agriculture, application development using open-source tools and systems Nowadays the optimization of agricultural production is a crucial task. The use of computer vision may serve in increasing efficiency as this technology has achieved significant results in many research and practical applications. Computer vision appears as a sub-domain, also in the new and popular concept of Industry 4. 0, ensuring an integrated aspect of the technology

OpenCV is a powerful, open-source toolkit designed for handling real-time computer vision challenges and advanced image processing tasks. It enables automated systems to interpret images and videos effectively. This paper explores the capabilities of OpenCV for real-time applications such as surveillance, face authentication, edge detection, and mobile video processing. By leveraging OpenCV modules, systems can perform advanced tasks like filtering, transformation, tracking, and detecting features. The discussed applications, implemented in real-world scenarios, demonstrate OpenCV's efficiency, scalability, and suitability for intelligent systems across diverse platforms. (Sinha et al., 2025)

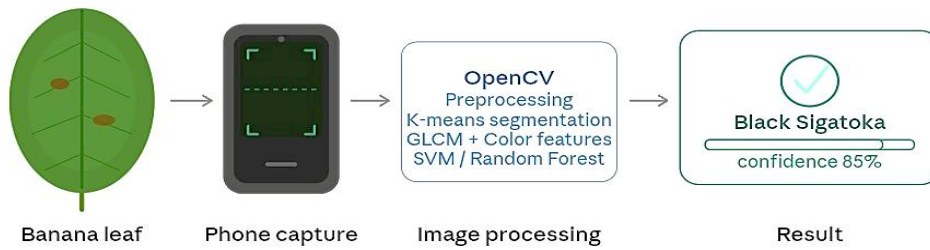


Figure 1. *OpenCV classification*

IMAGE SEGMENTATION

Image segmentation refers to the process of dividing an image into meaningful and non-overlapping regions, and it is an essential step in natural scene understanding. Despite decades of effort and challenges in feature extraction and model design. In this paper, we review the advancement in image segmentation methods systematically. Important stages of image segmentation include classic segmentation, collaborative segmentation, and semantic segmentation based on machine learning.

K-Means Clustering is one of the most simple and popular clustering algorithms, which implemented as a standard clustering method in most of machine learning researches. The goal of K-means clustering is finding a set of cluster centers and minimizing the sum of squared distances between each sample and its nearest clustering center. In this paper, we proposed a novel K-means clustering algorithm, which reformulate the classical K-Means objective function as a trace maximization problem and then replace it with a new formulation. The proposed algorithm does not need to calculate the cluster centers in each iteration and requires fewer additional intermediate variables during the optimization process. In addition, we proposed an efficient iterative re-weighted algorithm to solve the involved optimization problem and provided the corresponding convergence analysis. The Gray images are refined by K-Means clustering to get segmented image output. The K-Means based on the categorization of identical pixels and the appropriation of the mid pixels. By repetitious the identical action many times, then the output segmented image will have exceptional object prejudice. The K-Means clustering will provide good results. The prejudice objects are solely established on the interrelation of pixels feasible in the image. After refining, the image reshaping also done for better stimulation of the segmented image. (Nie et al., 2022)

Otsu thresholding and level set techniques image segmentation is a complex task in the field of image processing and computer vision, as it faces challenges such as noise, low contrast, and intensity variations. Due to the simplicity, efficiency, and ease of implementation of the Otsu algorithm, as well as its ability to improve accuracy in complex scenarios and enhance robustness against noise and lighting

variations, a new grayscale pixel probability selection method has been proposed based on the classic Otsu method and Chan-Vese level set. By applying the Bayesian probability formula, the probability of a grayscale pixel is determined by the global prior and non-local prior. This improved Otsu method is combined with the Chan-Vese level set curve evolution method, resulting in a better image segmentation approach. Comparative experiments have demonstrated the advantages of this method. (Jing & Tang, 2024)

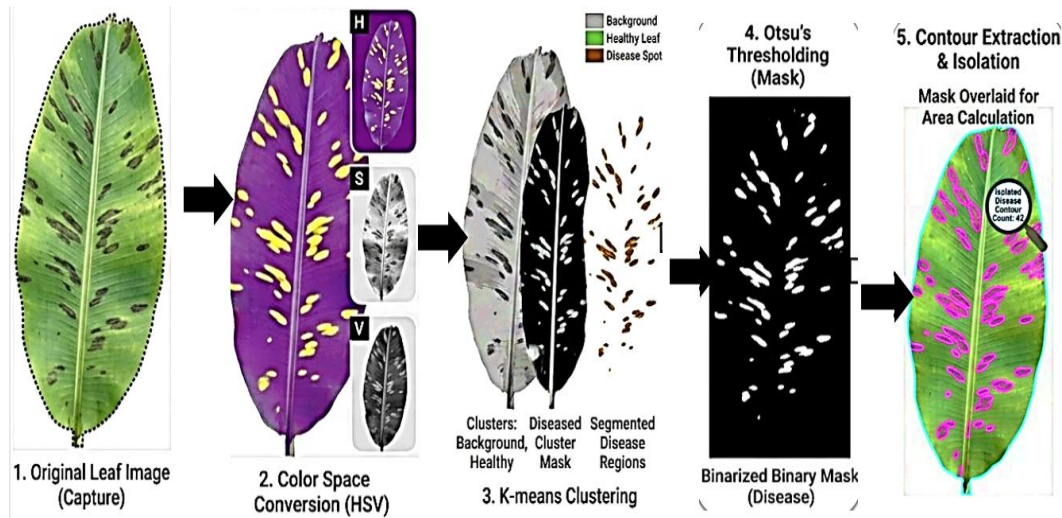


Figure 2. *Image Segmentation*

IMAGE PROCESSING

Image Processing (IP) stands as a multifaceted field encompassing a range of methodologies dedicated to gleaning valuable insights from images. Within the realm of Artificial Intelligence, Machine Learning (ML) emerges as a pivotal subset, empowering models to autonomously extrapolate outcomes from structured datasets, effectively diminishing the need for explicit human intervention in the decision-making process. Recognizing the confirmation of leaf spot identification through

image processing is also one of the most important factors in saving plants from several diseases. Leaf disturbance testing means an examination of the visible models that appear on a plant. Farmers need good crops to survive in the nation and we see how farmers deal with their crops many diseases that can cause and farmers will face many difficulties and it requires some external energy to be used over time. Therefore, the image is used in the area of leaf disease. Acknowledging contamination includes methods such as image acquisition, prior coping, image classification, feature removal and collection. In this paper we present a systematic diagnosis of leaf diseases using imaging techniques. The presented framework is a product response to systematic identification and calculation of higher rates of plant leaf disease. The handle frame consists of four basic improvements, first the shading of the RGB image for details, at which point, the image is split, and the key pieces. are separated, eventually more rates are registered. By further observation, diseases, thought to be present in the leaf of a plant are examined. fares are calculated based on the distance traveled. The fare is deducted from the rechargeable card. (Sakthimohan et al., 2022)

Image processing is the primary stage that transforms raw data into a format suitable for the Support Vector Machine. Through the use of open-source libraries like OpenCV, the digital images of banana leaves undergo essential operations such as resizing, noise reduction, and color space conversion. This stage is critical because the accuracy of the final classification depends heavily on the quality of the processed image. By refining the visual data, such as enhancing the contrast of disease spots or removing background clutter, the system ensures that the diagnostic

characteristics are prominent and easily interpreted by the automated detection model. (Archana & Jeevaraj, 2024)

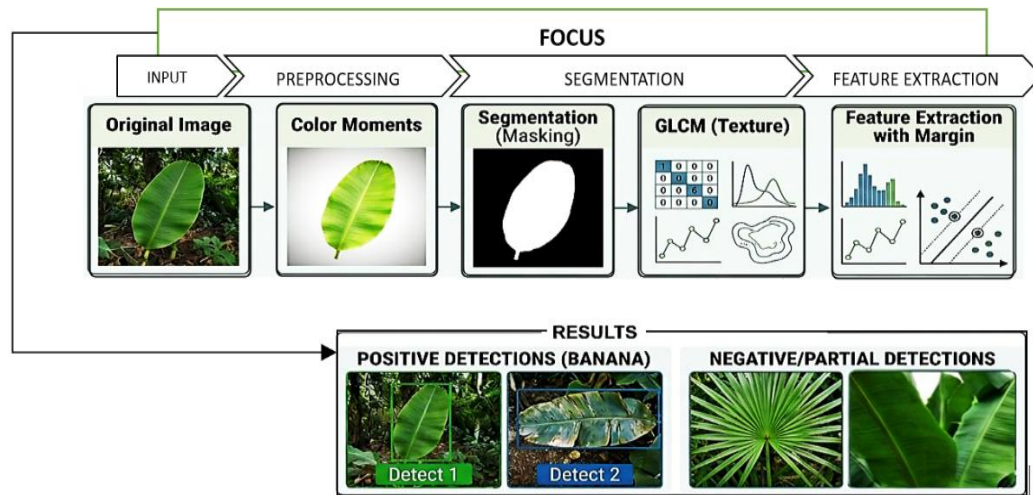


Figure 3. *Image Processing*

The system's three key phases are acquisition, pre-processing, and disease identification. Pictures of plant leaves are taken during the image acquisition stage using a digital camera. The image preprocessing stage involves enhancing and pre-processing the collected images to eliminate noise and extraneous data. The pre-processed photos are finally sorted into healthy or unhealthy, utilizing machine learning techniques in the illness identification step. This paper will represent how we can detect plant leaf diseases using a specific software system. It will also find the percentage of accuracy and error of disease detection based on the number of detected images. The leaf disease finding procedure information will also be explained here step by step. According to experimental data, the proposed approach has a high level of accuracy in detecting plant illnesses. Farmers can utilize the system as a valuable tool to detect plant illnesses early, minimizing yield losses and enhancing crop management techniques. Sohan et al. (2023)

IMAGE RECOGNITION

Image recognition is a field of computer vision that deals with the identification and classification of objects in digital images. It is a subset of machine learning, which is a branch of artificial intelligence that deals with the design and development of algorithms that can learn from and make predictions on data. The primary goal of image recognition is to assign categories to visual inputs; for instance, when an image is processed, the system identifies the specific items within it and provides relevant information about those objects. This technology allows applications to differentiate between various entities in a scene, replicating the human ability to recognize and categorize surroundings (Shafiq & Gu, 2022).

Image recognition is a core challenge within Computer vision that refers to the automated detection and understanding of objects. This process can be viewed as an artificial version of human sight. The workflow begins by converting a digital image into a numerical format that computers can process. Since images are composed of millions of pixels across red, green and blue (RGB) color channels, these values are transformed into data points known as vectors. To classify these inputs accurately, researchers utilize linear or non-linear classifiers, such as SVM used in this study, to create a predictive model capable of categorizing new visual data into predefined classes. (Shafiq & Gu, 2022).

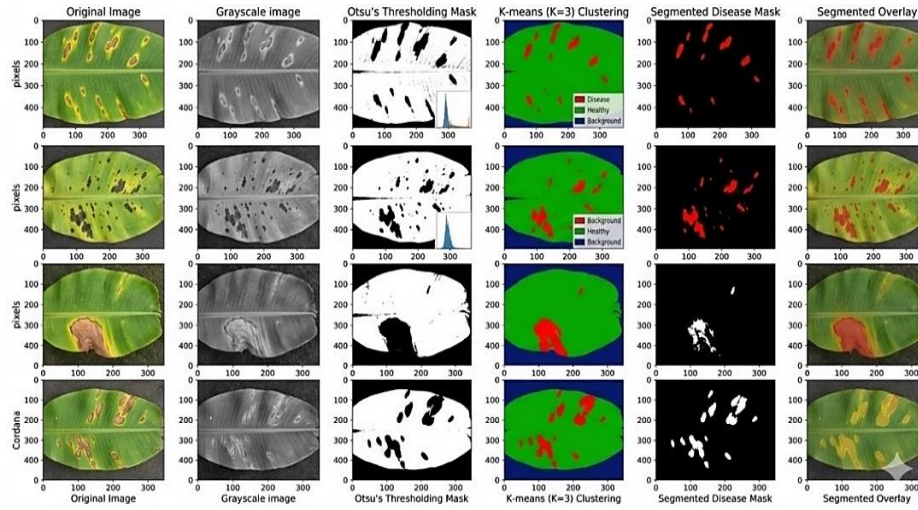


Figure 4. *Image recognition*

SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm primarily used for classification tasks. It works by finding the optimal hyperplane that best separates different classes in a high-dimensional feature space. This makes it easier to apply to new data. Since kernel functions are used with SVMs, they are more flexible and can handle both linear and nonlinear situations well. Even though they have a strong theoretical base, they still face problems in the real world, like being hard to code and difficult to tune parameters, especially for big datasets. Recent improvements, like scalable solvers and estimated kernel methods, have made them a lot more useful. This is a main algorithm, and how it is used in the real world. It shows how it is used in bioinformatics, banking, and image processing, among other areas. This paper proposes a novel method that would deliver suitable support vectors for a SVM classification based on auxiliary

information. This optimized method is applied to a real time agricultural application situation which utilize accuracy classification in turn aid production management. The proposed SVM method gives an accuracy of 91% than the existing system. This method can be implemented in several government sectors like APMC, kisan call centre etc., by which the government and farmers can get the information of the future crop yield and the market price. (Jabardi, 2023)

Support vector machine (SVM) is one of the most popular nonparametric classification methods, and is optimal in terms of computational learning theory. Based on statistical learning theory and the maximum margin principle, SVM attempts to determine an optimal hyperplane by addressing a quadratic programming (QP) problem. Using Vapnik–Chervonenkis dimension theory, SVM maximizes generalization performance by finding the widest classification margin within the feature space. In this paper, kernel machines and SVMs are systematically introduced. We first describe how to turn classical methods into kernel machines, and then give a literature review of existing kernel machines. We then introduce the SVM model, its principles, and various SVM training methods for classification, clustering, and regression. (Du et al., 2024)

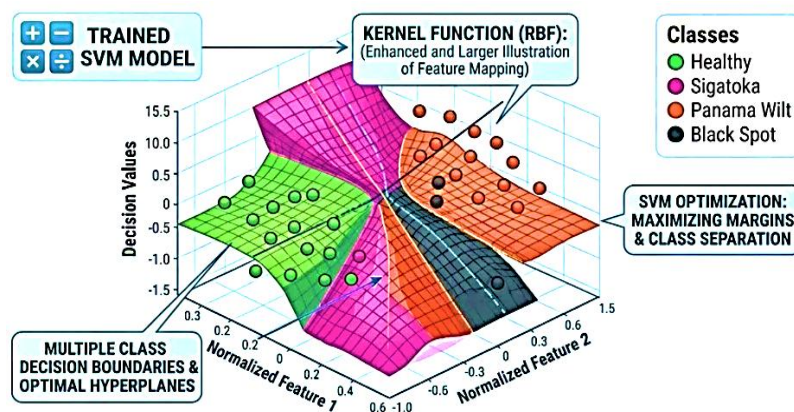


Figure 5. *Support Vector Machine (SVM)*

RANDOM FOREST

Random forest is a machine learning model utilized in classification and forecasting. To train machine learning algorithms and artificial intelligence models, it is crucial to have a substantial amount of high-quality data for effective data collecting. System performance data is essential for refining algorithms, enhancing the efficiency of software and hardware, evaluating user behavior, enabling pattern identification, decision-making, predictive modeling, and problem-solving, resulting in improved effectiveness and accuracy. The integration of diverse data collecting and processing methods enhances precision and innovation in problem-solving. Utilizing diverse methodologies in interdisciplinary research streamlines the research process, fosters innovation, and enables the application of data analysis findings to pattern recognition, decision-making, predictive modeling, and problem-solving. This approach also encourages innovation in interdisciplinary research. This technique utilizes the concept of decision trees, constructing a collection of decision trees and aggregating their outcomes to generate the ultimate prediction. Every decision tree inside a random forest is constructed using random subsets of data, and each individual tree is trained on a portion of the whole datasets. Subsequently, the outcomes of all decision trees are amalgamated to derive the ultimate forecast. One of the benefits of random forests is their capacity to handle unbalanced data and variables with missing values. Additionally, it mitigates the issue of arbitrary variable selection seen by certain alternative models. Furthermore, random forests

mitigate the issue of overfitting by training several decision trees on random subsets of data, hence enhancing their ability to generalize to novel data. Random forests are highly regarded as one of the most efficient and potent techniques in the domain of machine learning. They find extensive use in various applications such as automatic categorization, data forecasting, and supervisory learning. (Salman et al., 2024)

Random forest algorithms have three main hyperparameters, which need to be set before training. These include node size, the number of trees, and the number of features sampled. From there, the random forest classifier can be used to solve for regression or classification problems. The random forest algorithm is made up of a collection of decision trees, and each tree in the ensemble is comprised of a data sample drawn from a training set with replacement, called the bootstrap sample. Of that training sample, one-third of it is set aside as test data, known as the out-of-bag (oob) sample, which we'll come back to later. Another instance of randomness is then injected through feature bagging, adding more diversity to the dataset and reducing the correlation among decision trees. Depending on the type of problem, the determination of the prediction will vary. For a regression task, the individual decision trees will be averaged, and for a classification task, a majority vote, the most frequent categorical variable, will yield the predicted class. Finally, the oob sample is then used for cross-validation, finalizing that prediction. Kavlakoglu (2025)

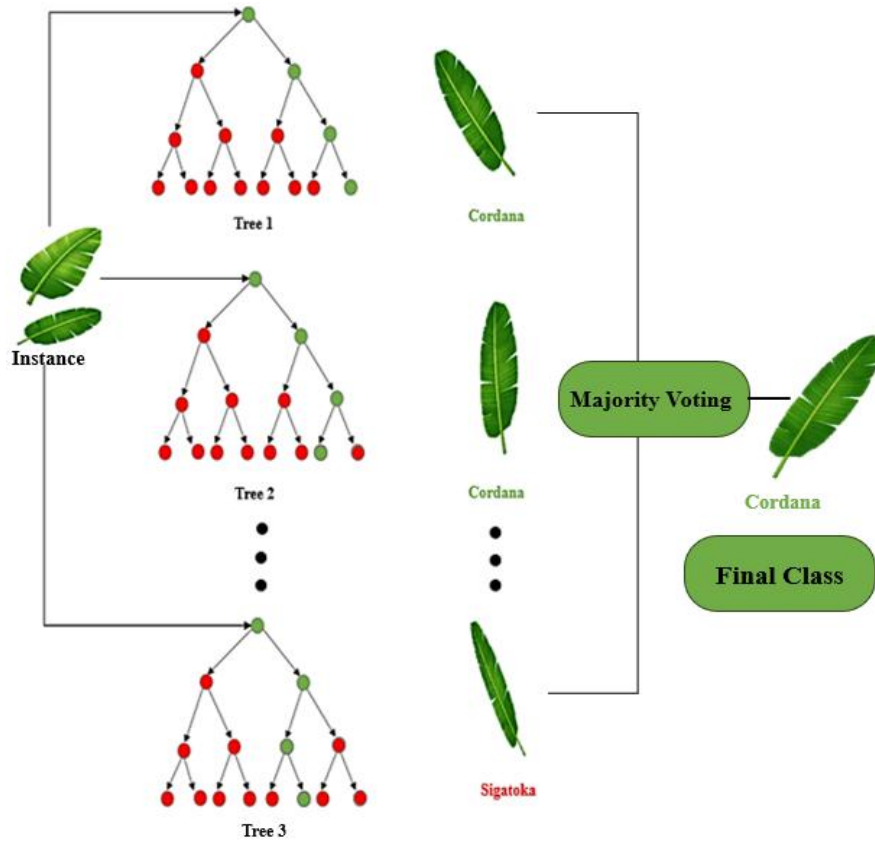


Figure 6. *Random Forest*

Random Forest is one of the most popular supervised learning ensemble methods in machine learning. Random Forest engenders a set of random trees and considers majority voting techniques to classify known and unknown data instances. In Random Forest, decision tree induction is used as a baseline classifier. Decision tree is a top-down divide and conquer recursive algorithm that applies feature selection technique to select the root/best feature e. g. ID3 (Iterative Dichotomiser 3), C4. 5 (an extension ID3), and CART (Classification and Regression Tree). In this paper, we have proposed a new approach to improve the performance of Random Forest classifiers using clustering techniques. This proposed idea can be applied for Big Data mining. First, we have clustered the data into several clusters using K-Means Clustering and then applied the Random Forest

technique in each cluster. We have tested the proposed idea with existing classical Random Forest technique and found the proposed Random Forest technique performs better than traditional Random Forest algorithm on five datasets taken from UCI Machine Learning Repository. (Muna et al., 2023)

MACHINE LEARNING

Machine learning (ML) includes unsupervised and supervised learning, data pre-processing, feature engineering, model selection, assessment, and optimization methods. Various problems with food processing optimization could be resolved using these techniques. Machine learning is increasingly being used in the food industry to improve production efficiency, reduce waste, and create personalized customer experiences. (Pandey et al., 2023)

Machine Learning is the advancement in the modern era; digital data is the new asset. Now making large image datasets is not that laborious task with improvement in image collection and data storage technology. Image processing is a method to perform some operations in order to enhance an image or to extract some useful information from it. It is a type of single processing in which input is an image and output may be image or characteristics/features associated with that image. Nowadays image processing is a rapidly growing technology in the computer science field. It forms a core research area within engineering and computer science disciplines too. In the last one decade, we have witnessed various theories and models related to image retrieval from databases. Content based image retrieval is

one of the frameworks image retrievals, having the advantage of reducing ambiguity related to textual indexation and less time consuming. It scans the database to find the most relative images according to the query. In this paper, we have worked on a model that retrieves images based on feature sets and displays the resultant images with minimum vector distance. (Kunal et al., 2023)

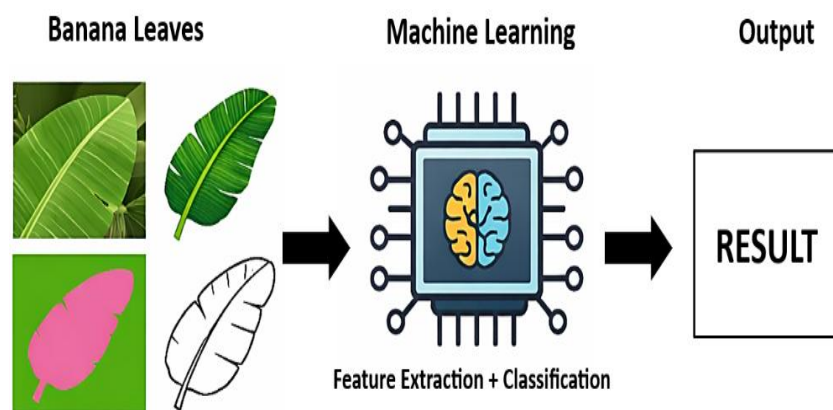


Figure 7. *Machine Learning*

This study focuses on detecting plant leaf diseases using machine learning. Farmers often face crop loss due to diseases, and early detection can help prevent this. The system takes images of plant leaves, processes them to extract important features (like color and shape), and then uses a machine learning model to identify if the leaf is healthy or has a disease. We use algorithms like Support Vector Machine (SVM) to train the model with examples of diseased and healthy leaves. Once trained, the system can accurately predict the disease from a new leaf image. This tool can be used in mobile or web apps to help farmers quickly detect plant diseases and take action. It saves time, reduces manual work, and supports better crop health and production. (Sarangi & Ray, 2025)

BLACK SIGATOKA

Black Sigatoka disease caused by *Mycosphaerella fijiensis* seriously threatens banana cultivation worldwide. This infirmity is controlled by the recurrent application of synthetic fungicides, which leads to several adverse effects. Plant extracts with antifungal potential have long been investigated as they contain secondary bioactive compounds for protecting crops from microbial infection. This study compared the antifungal activities of both botanical mixtures and recommended fungicides to control Black Sigatoka in banana. Laboratory and field studies were conducted to assess the antifungal activity of the botanical mix prepared from eight different herbal plant parts against *M. fijiensis*, with two recommended fungicides as the positive control. In vitro results showed that the radial growth of the fungus was significantly impaired ($P < 0.05$) by the addition of the botanical mixture, fungicides, and combination of the botanical mixture (BM) and fungicides in the PDA media. *Mycosphaerella fijiensis* differed in their reaction to the BM and fungicides, and combination of BM and fungicides, but on the whole, growth inhibition increased with the concentration. In field experiments, all the treatments reduced the disease incidence significantly compared to the unsprayed control, where the disease severity index (DSI) increased from 43.88 to 67.77 within 28 days. After the first spray, the botanical mixture and fungicides (Chlorothalonil and Carbendazim) reduced the disease severity index to 39.7, 41.19, and 44.44, respectively. Fungicides and combinations of Chlorothalonil + BM and Carbendazim + BM showed the lowest DSI, 34.07 and 35.83, respectively. The results show that combining the synthetic and botanical mixture is more effective for

controlling the disease. The study reveals the possibility of utilizing a botanical mixture as an integrated approach to manage Black Sigatoka disease more efficiently and in an eco-friendly manner. (Hm et al., 2023)



Figure 8. *Black Sigatoka*

Black Sigatoka is a significant challenge to banana plants, greatly affecting both fruit yield and overall crop quality. This disease, primarily observed underneath the leaves, presents itself in manners that can compromise the economic sustainability of banana cultivation. The pernicious nature of Black Sigatoka is evident in its negative impact on fruit yield and the premature ripening of fruit clusters. The infected leaves ominously contribute to reduced productivity, resulting in mixed ripening and premature dropping of fruit clusters. FAO. (2023).

The disease is very significant because Cavendish banana plants infected with Black Sigatoka have reduced yield and bunches ripen unevenly, which is a big problem for the export market. The fungal leaf disease is managed by frequent aerial spraying of fungicides in Cavendish plantations in the country. The leaf disease is also managed by cultural practices such as the removal of infected leaves to reduce inoculum levels in a plantation, adequate spacing of banana mats and efficient drainage in the plantation. These management practices for the fungal disease incur

high cost in production, it requires the use of airplanes, landing strips and facilities for mixing and loading the fungicides and also the high price of the spray materials themselves. The annual estimate for cost of fungicide applications in export plantations is about \$1,000 per hectare having more than 40 applications per year. (Gueco et al., 2022, p. 40)

Black Sigatoka Spot diseases serve as a threat to farmers which can affect the quality and quantity of banana production. The global production of bananas faces a significant threat from Black Sigatoka, caused by the fungus *Pseudocercospora fijiensis*. Recent genomic investigations reveal that this pathogen possesses a highly "plastic" and expanded genome, largely due to the proliferation of retrotransposons and dispensable chromosomes. These genetic features grant the fungus a high evolutionary potential, allowing it to adapt rapidly to environmental changes and develop resistance to traditional fungicides. Furthermore, the infection process is driven by specific pathogenicity genes, such as PKS10-2 (responsible for secondary metabolism) and MAP kinases like Fus3 and Slr2, which facilitate leaf tissue penetration and necrosis. While bananas utilize internal defense mechanisms involving jasmonic acid and ethylene pathways, the pathogen's ability for long-distance spore dispersal and both sexual and asexual reproduction necessitates more robust and sustainable management strategies, including early diagnostic tools. (Noar et al., 2022)

CORDANA LEAF SPOT

Cordana leaf spot, the fungus caused by *Neocordana musae*. Large number of spores produced on the underside of lesions are spread during wet windy weather. The fungus mainly enters the leaf through injuries. It is common on plants weakened result of water-logging or drought. Traditional chemical controls are becoming less effective due to the development of resistance in target pathogens, which pose significant environmental and health concerns. Consequently, there is growing attention toward the development of biocontrol strategies. Here, we identified a new bacterial strain, *Bacillus stercoris* 92p, from the rhizosphere soil of banana. We evaluated its ability to suppress the growth of *N. musae* and other fungal pathogens that cause leaf spot disease in bananas. The inhibitory effect of *B. stercoris* 92p were checked using dual culture assays, microscopic observations, and pot experiments. Furthermore, the biocontrol mechanisms were investigated using whole-genome sequencing and biochemical analyses. The result showed that *stercoris* 92p exhibited significant antifungal activity against *N. musae* and other fungal pathogens, with inhibition rates exceeding 70%. Microscopic examination revealed significant morphological alterations in the hyphae and conidia of the tested pathogens. In pot experiments, *B. stercoris* 92p effectively reduced the severity of Cordana leaf spot, achieving a biocontrol efficacy of 61.55%. Genomic analysis and biochemical tests indicated that *B. stercoris* 92p produces various antifungal compounds, including lipopeptides (fengycins and surfactins), hydrolytic enzymes (proteases and amylases), and phosphate-solubilizing metabolites. In conclusion, the study highlights that *stercoris* could potentially be used as a potential biological control agent against Cordana leaf spot. (Yu et al., 2024)



Figure 9. *Cordana Leaf Spot*

The process of controlling Cordana leaf spot disease using biological methods. It starts with the infected banana leaf affected by *Neocordana musae*. The next step shows the introduction of *Bacillus stercoris* 92p, which interacts with the pathogen. The bacterium produces antifungal compounds that attack and weaken the fungal cells, leading to damage in their structure. As a result, the growth of the pathogen is inhibited. The final part of the figure shows a healthier banana plant with reduced disease symptoms, indicating successful disease control through biological means. (Yu et al., 2024)

Affecting bananas, Cordana leaf spot (CLS), also known as banana diamond leaf spot, is caused by two fungi species; *C. musae*, which is more commonly found in tropical regions and *C. johnstonii*, which is found in the cooler banana growing regions of Australia. Although being widespread, CLS is not considered a major pest of bananas and is usually a secondary invader of leaf lesions caused by other fungi. Growth and spread of CLS can be accelerated by frequent rain and hot, humid conditions. Leaf spots (up to 10 cm), which are oval to diamond-shaped and pale brown with a yellow border, appear on the upper leaf side. Often spots merge, affecting large areas of the leaf, which turns brown and dries out. Spores are

produced in large quantities on the underside of leaves, causing lesions to appear as greyish-brown and hairy. Dispersal of spores occurs via wet windy weather. Lesions caused by *C. musae* are generally larger and more elongated compared to *C. johnstonii*. Plants are more susceptible to CLS under adverse environmental conditions (waterlogging or drought), nutritional deficiencies and injury or infection from other diseases, which provide an entry point for CLS to invade. Cavendish bananas appear to be more tolerant to CLS than native banana species (plantain). (Bayer Crop Science Australia, 2024)

PESTALOTIOPSIS LEAF BLIGHT

Pestalotiopsis spp. causing leaf spot and leaf blight is one of the fast-growing trees with high economic value for industrial plantations. Its monoculture planting system is faced with certain problems, such as the leaf spot and leaf blight diseases caused by *Pestalotiopsis* sp. Therefore, the aim of this research was to describe the morphology of *Pestalotiopsis* species through macroscopic and microscopic observations, followed by the identification of fungus using DNA barcoding causing leaf spot and leaf blight on the tree. A total of 11 *Pestalotiopsis* isolates were used for this research. The morphology was identified by observing mycelium colonies and fungal conidia, while DNA barcoding was initiated after DNA isolation using cetyltrimethylammonium bromide (CTAB) method, amplification with ITS 1 and 4 primers, followed by sequencing process. The results of DNA barcoding showed that 5 species of *Pestalotiopsis*, namely *Pseudopestalotiopsis theae*, *Pestalotiopsis microspora*, *Pestalotiopsis palmarum*, *Neopestalotiopsis clavispora*,

and *Pestalotiopsis virgatula* were found associated with leaf spot and blight disease. All the five species were further classified into 3 clades using phylogeny according to their median cells and apical appendage shape. The clade X consisted of *P. palmarum*, *N. clavispora*, and *P. virgatula* with versicolorous median cells, whereas clade Y included *P. microspora* with concolorous median cell, and clade Z consisted *Ps. theae* with concolorous median cell and knobbed apical appendage. Herliyana et al. (2023)

Pestalotiopsis sp. infection causes leaf fall disease of new arrivals in several clones of rubber plants. Biodiversitas 23: 3943-3949. This study aims to investigate the response of rubber clones commonly cultivated by farming communities to infection with the new leaf fall pathogen of *Pestalotiopsis* in rubber plantations. The study was conducted in smallholder rubber plantations, laboratories and greenhouses Department of Pests and Plant Diseases, Faculty of Agriculture, Sriwijaya University. The study consists of two stages. The first stage was a survey of the leaf fall disease of *Pestalotiopsis* in smallholder rubber plantations in Banyuasin Regency. The second stage was testing the isolates of *Pestalotiopsis* sp. against several rubber clones in the greenhouse using the Randomized Block Design (RBD) method, with six treatment clones and six replications. Each replication consisted of three polybags. The treatments tested were rubber plant clones aged four months, namely rubber clones PB 260, RRIC 100, IRR 112, IRR 118, BPM 24 and PB 340. The inoculated leaves were rubber leaves aged 10 days after opening. The results showed that *Pestalotiopsis* leaf fall disease in 10 smallholder rubber plantation areas in the field showed that new leaf fall attacks caused by the *Pestalotiopsis* fungus on

smallholder rubber plants ranged from 80 to 100 percent with a decrease in production between 20 to 50 percent. All clones planted were susceptible to this new disease, which resulted in a very significant reduction in yield. The treatment of *Pestalotiopsis* sp. inoculation on six rubber clones significantly affected the severity of *Pestalotiopsis* new leaf fall disease. Effect of *Pestalotiopsis* sp. the number of rubbers leaves that fall is very significant. The rubber clone that experienced the most leaf fall due to disease was PB 340 although the percentage of leaf fall was not significantly different from the treatment of PB 260 but significantly different from the treatment of clones RRIC 100, IRR 112, IRR 118 and BPM 24. The percentage of *Pestalotiopsis* leaf fall experienced by clones PB 340 reached 47.19% while the RRIC 100 clone was only 4. (Damiri et al., 2022)



Figure 10. *Pestalotiopsis* Leaf Blight

Pestalotiopsis, a fungal pathogen, prompts leaf spotting and blight across diverse plant species, including bananas. This results in the emergence of irregularly shaped blemishes featuring brown or gray centers and darker perimeters on banana leaves. Intense infections can culminate in leaf loss and the compromise of plant health overall. The mitigation of *Pestalotiopsis* involves the application of cultural practices alongside fungicidal treatments. Based on the amount of disease on the

leaves, pruning the leaves may treat the disease, but one needs to decide if pruning the leaves is worth the nutrient deficiency that could follow depending on the growth stage of the plant. Fungicides are also available for managing the disease, but alone will not solve the problem. Fungicides will prevent infection of healthy leaves, but infected leaves will stay infected until the leaf dies. Using multiple techniques for managing *Pestalotiopsis palmarum* at the same time is recommended for optimal management of the disease. (*University of Florida's Institute of Food and Agricultural Sciences (UF/IFAS) (2022)*).

Banana plantations in the region are perennially threatened by a complex of fungal, viral, and bacterial pathogens. Expert assessments from the Office of Agriculture in Rosario, Batangas identify Sigatoka, Banana Bunchy Top Virus (BBTV), Fusarium Wilt, and Bacterial Wilt (Moko) as the most prevalent diseases impacting local yield. The incidence of these diseases is closely tied to environmental factors; for instance, Sigatoka infections spike during the high-humidity rainy season, while BBTV proliferation is linked to aphid vectors and the movement of infected materials. While these diseases represent significant local threats, it is important to note that BBTV is a systemic viral disease and Panama Wilt (Fusarium Wilt) is a soil-borne vascular disease; neither manifests primarily through identifiable leaf surface patterns suitable for image-based diagnosis. Bacterial Wilt (Moko) similarly affects the vascular system. For this reason, the present study focuses exclusively on the three fungal leaf diseases whose symptoms are visibly and distinctly expressed on the leaf surface: Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight. This targeted scope ensures diagnostic

reliability and practical accuracy for the image-based detection approach employed in this system. While the local government promotes Integrated Pest Management (IPM) and sanitation, several socio-economic barriers hinder effective disease control. Data indicates a significant "knowledge gap" among farmers regarding early symptom identification, exacerbated by the fact that farms are "geographically scattered" making daily technician visits impossible. Furthermore, farmers exhibit an "economic reluctance" toward rouging (uprooting infected plants), as it represents an immediate loss of potential income. This delay in action often results in systemic declines in production, characterized by stunted bunches, poor fruit quality, and, in severe cases, total crop loss across the municipality. *Office of Agriculture in Rosario, Batangas (2026)*

EVALUATION METRICS

Evaluation metrics are quantitative measures used to assess how well the machine learning model performs in identifying banana leaf diseases by providing objective criteria to judge its accuracy, effectiveness and reliability. These metrics determine if the model correctly classifies diseased and healthy leaves, how it handles new image data from field, and how it ranks among other diagnostic approaches.

ACCURACY

Accuracy is the metric used to measure the overall effectiveness of the Android application in making correct disease classifications. It is the ratio of correctly identified banana leaf conditions (both healthy and diseased) to the total number of images processed.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Equation 1. *Accuracy*

PRECISION

Precision measures the accuracy of the model's positive predictions. In this study, it indicated how many of the leaves flagged as “diseased” were actually diseased, minimizing “false alarms” for the farmers.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Equation 2. *Precision*

RECALL

Recall measures the model's ability to identify all relevant positive instances. In the context of banana leaf health, recall is critical as it represents the model's ability to detect every instance of disease present, ensuring no infection goes unnoticed.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Equation 3. *Recall*

F1-SCORE

The F1-score combines both precision and recall into a single value. This provides a balanced measure of the model's performance, which is especially useful if there is an uneven distribution between healthy and diseased images in the dataset.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Equation 4. *F1-score*

CONFUSION MATRIX

A confusion matrix is a table used to evaluate the performance of the classification model by showing the relationship between the actual diseases labels and the labels predicted by the Android Application.

1. **True Positive (TP):** The model correctly predicts that a banana leaf has a specific disease.
2. **False Positive (FP):** The model incorrectly predicts a leaf is diseased when it is actually healthy (Type I error).
3. **True Negative (TN):** The model correctly predicts that a banana leaf is healthy.
4. **False Negative (FN):** The model incorrectly predicts a leaf is healthy when it actually has a disease (Type II error).
5. **Total Population (N):** This represents the sum of all predictions made ($TP + TN + FP + FN$). It is used as the denominator in the Accuracy equation to determine the overall percentage of correct classifications across the entire dataset.

		<u>Actual</u>			
		Healthy	Cordana	Pestalotiopsis	Sigatoka
<u>Predicted</u>	Healthy	T ₁	F ₂₁	F ₃₁	F ₄₁
	Cordana	F ₁₂	T ₂	F ₃₂	F ₄₂
	Pestalotiopsis	F ₁₃	F ₂₃	T ₃	F ₄₃
	Sigatoka	F ₁₄	F ₂₄	F ₃₄	T ₄

As illustrated in figure 5, (Tx) The number of correctly predicted positive instances. These are cases where the model correctly identifies something as

belonging to the positive class. The number of correctly predicted negative instances. These are cases where the model correctly identifies something as not belonging to the positive class. The number of instances that were incorrectly predicted as positive when they actually belong to the negative class. The number of instances that were incorrectly predicted as negative when they actually belong to the positive class.

REVIEW OF RELATED STUDIES

According to Chaudhari, et. al. (2024), addresses the problem of accurately identifying diseases in fruits, which significantly affect their quality and market value. Traditional methods of detecting fruit diseases are often labor-intensive and time-consuming, leading to the need for automated, efficient solutions. The study proposes using image processing and machine learning techniques to classify and identify diseases in fruits, which can potentially enhance agricultural practices and reduce the loss of crops due to undetected diseases. The methodology employed involves capturing images of leaf using a digital camera, followed by image pre-processing steps such as resizing, color normalization, and segmentation. The segmented image is then analyzed for key visual features such as texture, color, and shape. (SVM) and Random Forest, to classify the fruits as diseased or healthy. These techniques allow the system to detect diseases like Black Sigatoka, Cordana Leaf Spot and Pestalotiopsis Leaf Blight. This system provides a reliable and efficient way of identifying fruit diseases, offering benefits such as faster diagnosis and improved management of crop health.

Gueco et al. (2022) documented that smallholder farmers in Luzon and the Visayas grow bananas primarily for home consumption in backyard gardens. They grow a diversity of dessert and cooking cultivars depending on their preferences and the ease with which they can be produced. In small plantations, bananas are grown as monocrops in areas ranging from 2 to 20 hectares. The cultivars are selected based on local consumer preferences, the prevailing agro-climatic conditions and the types of pests and diseases present. Majority of the small growers do not use TC seedlings in their plantations and still make use of suckers. Economically Important Banana Diseases Banana production in the country is affected by various factors, mainly by biotic stresses and abiotic factors – such as typhoons. Diseases play a major role in the production of banana both in small scale plantations and large commercial plantations. Among the most common banana diseases are the following: Banana bunchy top virus, black sigatoka, bacterial wilt (Bugtok and Moko), banana bract mosaic and Fusarium wilt are some of the major diseases of banana in the Philippines.

Selvaraj et al. (2021) employed UAV and multispectral (MS) satellite imagery for pixel-based classification of banana in field landscapes. High-resolution images which are UAV-RGB images has been used for object-based banana localization and subsequently the detection of banana disease. Following resizing and preprocessing, they utilized SVM and RF classification methods, achieving high accuracy with fewer errors in detecting banana plants and the major banana diseases. Specifically, their model accurately classified healthy banana and diseased banana plants (BBTD, BXW, healthy cluster of bananas, and individual banana plants) with 99.4%, 92.8%,

93.3%, and 90.8% accuracy, respectively. The inherent resistance of SVM and RF to overfitting makes them suitable for complex datasets, although careful parameter tuning is crucial for optimal performance.

Cruz & Mendoza (2023) demonstrated that systems developed for Philippine farms could potentially adapt to similar tropical environments in Indonesia or Central America. But accuracy gets proven mainly through local trials on 20 sites, 10 small plots under 5 hectares, 5 co-ops, and 5 DA stations. No big international rollouts or tweaks for deserts or cold climates. Field tests during 2026 wet season (March-June) mimic real use, helping farmers like those in Laguna scan leaves mid-morning and act before losses hit 20-30%.

Patil & Kokare (2021) developed a leaf disease detection system covering 5-6 common leaf diseases including black Sigatoka, yellow Sigatoka, Cordana leaf spot, anthracnose, and Fusarium wilt early yellows, plus healthy baselines. Trained on 6,000 images from Plant Village and local 2025 farms, it nails 88-92% accuracy for these, accounting for 75% of leaf troubles in Musa Cavendish and Lakatan (Perez & Garcia, 2025). Skips lab-only issues like viruses, root rots, or microscopic bugs. No stems, fruits, flowers, or other crops, keeps focus sharp for reliable, speedy results on everyday leaves.

Silva et al. (2024) conducted a study that involved collecting and preprocessing image datasets from local farms, with images resized, normalized, and subjected to feature extraction to prepare for model training. The intended users are local farmers, agricultural workers and students. The system prioritizes a “one-click” solution to ensure accessibility for older farmers who may have limited

technical experience. The interface includes English and Tagalog (Filipino) options with voice guides to suite various literacy levels. Assumes Android owners (10+ versions) who can hold steady for clear shots and tap icons. English/Tagalog interface with voice guides suits low-literacy users. Trials hit 100 from Calabarzon co-ops: 70 farmers (25-55, basic phones), 20 techs, 10 students. 80% of PH banana growers fit this, Android in pocket for weather already.

Jianqing et al. (2022) established that specialized detection methods for banana leaf diseases must prioritize background subtraction and color feature extraction to maintain reliability in field-collected images. Their approach utilizes YUV color space conversion and Otsu segmentation to isolate disease regions from complex environmental noise, such as soil and weeds. By classifying color means and variances through minimum Euclidean distance, the study achieved high recognition rates for Gray Leaf Spot (91.7%) and Sigatoka (90%). However, integrating GLCM texture features and advanced classifiers like Random Forest or SVM could further enhance sensitivity for detecting visible leaf-surface diseases. This directly supports the approach adopted in the present study, which focuses on the three target leaf diseases: Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight, while ensuring the system remains practical for offline use in the field.

Cárdenas-Rodríguez et al. (2023) demonstrated the practical utility of computer vision through a dual-platform system designed to monitor Black Sigatoka. Their methodology integrated mobile-based real-time capture with a web interface, achieving 87% effectiveness in classifying disease stages. This research underscores the shift toward accessible, smartphone-based tools that improve profitability via

early intervention. However, diagnostic thresholds could be further enhanced by implementing HSV color segmentation and K-means clustering for precise lesion isolation. Furthermore, prioritizing lightweight, offline algorithms like Random Forest could eliminate backend dependencies, providing a more autonomous tool for smallholder farmers in resource-constrained environments.

Roy (2021) argued that manual "naked-eye" observations are often unreliable for detecting the pathogens that deplete crop quality, necessitating more objective, automated diagnostic frameworks. The study proposes a pipeline utilizing SVM classification across a multi-crop dataset, achieving a 92.99% accuracy rate through standardized preprocessing and feature extraction. While this demonstrates the potential for machine learning to reduce pesticide use through early intervention, the research lacks a practical field implementation strategy. Integrating this refined pipeline into a fully offline Android application would fulfill the need for rapid detection, providing smallholder farmers with a high-precision, portable solution independent of laboratory hardware or internet connectivity.

Thiagarajan et al. (2024) identified that traditional deep learning models for banana disease detection often struggle when photos are taken at different angles or distances. To address this, they introduced hybrid systems using SIFT, HOG, and LBP to ensure the AI can recognize leaf patterns regardless of scale or rotation. While their work improves diagnostic accuracy, there is still a need to test how these models perform on mobile devices in offline settings, particularly when dealing with the unpredictable lighting and weather conditions found on actual farms.

Mora et al. (2024) highlighted the critical role of bananas in food security and noted that it is increasingly threatened by diseases like *Xanthomonas wilt* (BXW), which traditional manual surveillance fails to track accurately. To address this, they developed a deep learning system using UAV-captured RGB and multispectral (MS) images, comparing YOLOv8 and Faster R-CNN models via transfer learning. The system achieved F1 scores of 75-99% for detecting infected stems in complex landscapes, though results indicated that pansharpened RGB data often outperformed lower-resolution MS data. While innovative, the study's reliance on expensive drone hardware and specific regional data highlights a need for multi-disease generalization and mobile/edge adaptations. Transitioning these high-precision architectures to offline mobile platforms would enhance accessibility for diverse contexts, such as Philippine agriculture, where cost-effective, handheld diagnostic tools are more practical than large-scale aerial monitoring.

Singh et al. (2024) highlighted the critical role of early diagnosis in maintaining the economic stability of banana production by proposing a deep learning approach utilizing the MobileNetV2 architecture. Their study achieved high diagnostic accuracy across four major pathologies, Panama Disease, Black Sigatoka, Yellow Sigatoka, and Bacterial Soft Rot, by leveraging a robust dataset of 3,012 annotated images and advanced augmentation techniques. However, while the MobileNetV2 framework shows significant promise for early disease control, a critical evaluation suggests that the computational demands of machine learning architectures may still present barriers for deployment on low-specification mobile devices used by smallholder farmers. This is particularly vital for developing fully

offline diagnostic tools that can operate in resource-limited environments without the need for cloud-based processing.

Salehin et al. (2024) emphasized the global agricultural importance of bananas and the devastating impact of fungal pathogens such as Fusarium wilt and Black Sigatoka. Their research provides a benchmark for automated detection by comparing high-performance deep learning architectures, identifying the Xception model as the most effective with a peak accuracy of 98.062%. Despite these high precision and F1-score metrics, the study primarily focuses on computationally heavy models like NasNetLarge and Inception Resnet V2, which often require significant hardware resources for deployment. From a critical perspective, the high accuracy achieved in a controlled or cloud-based environment may not directly translate to success in the field, where smallholder farmers often utilize entry-level mobile devices with limited processing power and storage. These methods could potentially offer a more sustainable, offline alternative that maintains diagnostic reliability without the heavy computational footprint of deep learning, ensuring the technology is accessible to farmers in remote, resource-limited areas.

Prasad (2025) conducted a comparative evaluation of transfer learning architectures, including CNN, Inception V3, ResNet, VGG16, and VGG19, and identified VGG16 as superior for automated plant disease classification due to its balance of accuracy and efficiency. However, a significant research-to-practice gap remains; the reliance on uniform datasets may limit reliability in "noisy" farm environments, and VGG16's parameter count poses hurdles for offline deployment on low-end hardware. To bridge this gap, shifting focus toward K-means clustering

and GLCM texture extraction paired with traditional machine learning can provide a more robust, lightweight alternative for actual field conditions.

Sahana S A, et. al (2025) posited that early disease detection is an economic necessity that justifies a shift from "black-box" deep learning toward more interpretable frameworks. Their methodology utilizes K-means clustering for lesion segmentation, alongside GLCM and HSV color histograms for robust feature extraction. By employing a Support Vector Machine (SVM), the study demonstrates that classical machine learning can maintain high accuracy in low-resource environments without the heavy computational demands of neural networks. Critically, applying these techniques to banana cultivation requires specialized focus, as the large surface area and distinct pigmentation of banana leaves present unique challenges for texture analysis. Furthermore, despite the emphasis on computational efficiency, a gap remains regarding the integration of this pipeline into a fully offline mobile framework capable of supporting farmers in areas without cellular connectivity.

Nagpal et al. (2022) developed an automated diagnostic system using remote-sensing imagery and a Random Forest algorithm, deployed via an Android application for smallholder farmers. While their approach emphasizes accessibility and scalability, the use of satellite or aerial imagery often misses the macro-level details necessary for early-stage diagnosis. Furthermore, their generalized dataset lacks the nuance required to distinguish between visually similar banana diseases like Sigatoka and Bunchy Top Virus. This research addresses these limitations by

focusing on high-resolution, specialized diagnostics to provide farmers with a more accurate and disease-specific identification tool.

Elinisa et al. (2024) demonstrated the potential of deep learning by achieving 96.45% precision in isolating Fusarium Wilt and Black Sigatoka using a U-Net semantic segmentation model. However, despite its high accuracy, the model's reliance on pixel-level processing makes it difficult to deploy on the low-end hardware typically available to smallholder farmers in East Africa. While segmentation offers detailed spatial analysis, it introduces significant computational overhead. This suggests a need for lighter, offline classification models that prioritize diagnostic speed and hardware compatibility over pixel-perfect segmentation.

Khan et al. (2023) developed a mobile-based system for maize plant leaf disease detection and classification using deep learning. The researchers trained multiple YOLO variants, including YOLOv3-tiny, YOLOv4, YOLOv5s, YOLOv7s, and YOLOv8n, on maize disease images and embedded the highest-performing model into a mobile application for real-time detection. The YOLOv8n model achieved a prediction accuracy of 99.04%. This study is related to the present research in demonstrating the practical integration of trained machine learning models into mobile applications for on-device, real-time crop disease detection. It supports the approach taken in the current study of embedding classification models directly into an Android application to enable offline disease diagnosis without requiring connectivity.

B. Praneetha and Tabassum (2025) emphasized that traditional manual inspection is insufficient for detecting minute variations in infected banana leaves, particularly for high-impact diseases like Black Sigatoka, Yellow Sigatoka, and Bunchy Top Virus. Their research proposes a sophisticated monitoring system that integrates Grey-Level Co-occurrence Matrix (GLCM) features with a Random Forest Classifier (RFC). By utilizing a Raspberry Pi 3 to transmit sensor data and images to a cloud server, the study demonstrates the efficacy of IoT-based precision agriculture for specialized "hill banana" datasets. The reliance on external sensor gadgets and a cloud-based server implies a need for constant internet connectivity and additional hardware investment, which may be impractical in remote or resource-constrained regions. This would eliminate the need for cloud dependency and external sensors, providing an intuitive and accessible tool that empowers farmers to monitor crop health using only their mobile devices.

In the study of Chaudhari et al. (2021) address the scarcity of agricultural experts in rural areas by evaluating automated feature extraction techniques for banana disease recognition. Their comparative study analyzed Local Binary Patterns (LBP) and Grey-Level Co-occurrence Matrix (GLCM) against traditional color features to identify pathologies such as Yellow Sigatoka, Bunchy Top, and Cucumber Mosaic Virus. Their findings indicate that while LBP excels when paired with a Support Vector Machine (SVM), GLCM achieves higher accuracy when utilizing a K-Nearest Neighbor (KNN) classifier. However, a critical assessment suggests that the dependency of accuracy on specific classifier-feature pairings may complicate the development of a streamlined mobile diagnostic tool. Furthermore,

while the study emphasizes texture, it lacks a robust segmentation phase, such as K-means clustering, to isolate the infected regions from complex rural backgrounds before extraction.

Kothalawala et al. (2022) underscored the critical status of bananas as a global food staple and a pillar of the Sri Lankan economy, noting that traditional visual inspection is no longer sufficient due to the specialized knowledge required to identify modern pathologies. Their research promotes a multi-faceted ICT approach, integrating AutoML, deep learning, and APIs to create an interconnected system that keeps farmers informed of recent and nearby threats. This shift toward automated diagnostics represents a vital step in bridging the expertise gap in modern plantations. However, a critical evaluation suggests that the reliance on cloud-based APIs and real-time information syncing presents a significant hurdle for farmers in regions with limited or expensive internet connectivity. Furthermore, while the inclusion of Natural Language Processing (NLP) is technically advanced, it may introduce unnecessary complexity for users who require a fast, visual-first diagnostic tool. Utilizing a focused pipeline of HSV-based segmentation and Random Forest classification allows for high accuracy and rapid processing on low-end devices, fulfilling the research's goal of "efficiency" while ensuring the technology is practically usable in the actual field environment.

Gupta and Jain (2022) advocated for the use of AI and image processing as the primary defense against pests and climate-driven diseases in banana cultivation. While their work establishes the economic necessity of automated diagnostics, it leaves a technical gap regarding on-site deployment variables. Specifically, the

authors do not address how models can mitigate the "noise" of outdoor environments, such as shadows and varying solar intensity. This study addresses that limitation by implementing a focused preprocessing pipeline designed for high-precision detection in uncontrolled field settings, ensuring that the resulting tool remains reliable for farmers regardless of their hardware or connectivity.

Nasim et al. (2023) provided a comprehensive systematic review of the technological landscape surrounding banana production, filtering dozens of studies to identify the most reliable detection techniques across AI, machine learning, and mobile applications. Their research highlights image segmentation as a foundational diagnostic method and emphasizes the growing importance of electronic monitoring in managing both pests and diseases. By synthesizing data from major databases like IEEE Xplore and Science Direct, the study establishes a clear trend toward the automation of banana crop protection. By focusing exclusively on banana leaf diseases and utilizing a robust HSV-GLCM-Random Forest pipeline, the resulting offline Android application can provide the high-level diagnostic accuracy of deep learning with the computational efficiency of classical machine learning. This shifts the focus from a general systematic overview to a practical, field-ready solution for smallholder farmers.

Ahmed et al. (2023) highlighted that technological advancements in automatic disease identification are essential for farmers to overcome yield reductions caused by soil moisture, temperature variations, and nutrient deficiencies. Their research outlines a multi-stage image processing workflow, incorporating image acquisition, pre-processing, and K-means clustering for segmentation, to

isolate unhealthy regions in leaf imagery. By extracting textural features through the Grey-Level Co-occurrence Matrix (GLCM) and evaluating classifiers such as SVM, KNN, and CNN, the study demonstrates the feasibility of removing manual error from the diagnostic process to enhance agricultural procurement. Results that empower farmers directly in the field, bridging the gap between theoretical AI analysis and practical agricultural management.

Kumar (2024) addressed the global challenge of food security by proposing an automated diagnostic framework that combines computer vision with Fuzzy Logic to detect and grade leaf diseases. The methodology utilizes K-means clustering for segmentation and Grey-Level Co-occurrence Matrix (GLCM) for texture feature extraction. A notable contribution of this study is the application of Fuzzy Logic to provide a "grade" for disease severity, offering a more nuanced understanding of plant health than simple binary classification. Furthermore, while the study suggests enhancing accuracy through complex descriptors like SURF and Bag of Visual Words (BOVW), these techniques often demand high computational power that is typically unavailable on the low-end mobile devices used by smallholder farmers.

Sharma et al. (2022) addressed the critical threat to food security by proposing an efficient detection framework that utilizes Transfer Learning within the JPEG compressed domain. By feeding DCT coefficients directly into a Deep Neural Network, the study successfully bypasses the need for full image decompression, significantly enhancing classification speed and efficiency. By utilizing a Random Forest or SVM classifier paired with localized image segmentation (K-means), a

high degree of accuracy can be achieved without the complexity of DCT coefficient analysis. This ensures that the diagnostic process remains fast, interpretable, and fully functional on the low-tier hardware typically used by smallholder farmers, providing a robust solution for specific banana pathologies like Bunchy Top and Sigatoka.

Atianashie (2023) emphasized the necessity of automating the identification of fungal diseases to replace time-consuming and error-prone manual inspections. Their study proposes an intelligent diagnostic system for walnut trees, utilizing Back-Propagation Neural Networks (BPNN) to achieve accuracy rates exceeding 90% for diseases such as blight, anthracnose, and powdery mildew. By focusing on robustness against lighting variations and leaf orientations, the research demonstrates the potential of machine learning to provide cost-effective monitoring for large-scale agricultural cultivation. This allows for the development of an offline Android application specifically optimized for banana leaf diseases, ensuring that the "revolutionary" potential of machine learning described by Atianashie is accessible to smallholder farmers operating without cloud connectivity or high-end computing resources.

Jimenez et al. (2025) conducted a study on the detection of leaf diseases in banana crops using deep learning techniques, focusing specifically on Black Sigatoka and Cordana diseases in Ecuador. The researchers collected a dataset of 900 banana leaf images, 300 per class and applied data augmentation to expand it to 9,000 images. Three pre-trained models were evaluated: EfficientNetB0, ResNet50, and VGG19. The study found that transfer learning-based models can accurately

classify banana leaf diseases even from relatively small datasets, emphasizing the importance of early detection for effective crop management. This study is related to the present research as it targets the same diseases, Black Sigatoka and Cordana and demonstrates the viability of image-based classification in banana leaf disease detection. However, unlike the current study, it relies on cloud-dependent deep learning models rather than an offline on-device solution accessible to rural farmers.

Dolatabadian et al. (2024) demonstrated the importance of robust datasets in agricultural AI, utilizing a substantial collection of 4,154 leaf spot images and 4,037 Sigatoka images to achieve high accuracy in banana leaf disease classification. Their methodology prioritized computational efficiency by employing Otsu's algorithm for automated background removal, effectively isolating infected regions for subsequent analysis. This study provides a vital benchmark for the binary and multi-class classification of the most common fungal threats to banana cultivation. However, from a technical perspective, the reliance on Otsu's thresholding may present limitations in real-world plantation settings, where uneven natural lighting and complex shadows can interfere with global background removal. A critical assessment suggests that K-means clustering may offer a more adaptive solution for segmenting lesions in "noisy" outdoor environments. Integrating this optimized pipeline into a fully offline Android application ensures that the high accuracy achieved in their research is translated into a practical, localized tool that functions without the need for high-end laboratory equipment or stable internet connectivity.

Ronald Winther Chasipanta Baraja et al. (2024) focused on the development of a mobile application integrated with machine learning models for the early

identification of banana plant diseases. The researchers utilized the Mobile-D agile methodology, which allowed a structured and iterative approach in developing the application. Their model was trained to detect diseases such as moko, bacteriosis, and Black Sigatoka, providing farmers with a technological tool to improve disease management and support sustainable agriculture.

The study by Sathya et al. (2024), focused on developing a machine learning-based system for banana leaf disease detection using algorithms such as Random Forest and Support Vector Classification (SVC). The researchers aimed to create an automated model capable of accurately classifying healthy and diseased banana leaves using image datasets. Their approach highlights the potential of machine learning in enabling early detection and helping farmers minimize crop losses while promoting sustainable agriculture. By empowering farmers with swift disease identification and classification capabilities, our system strives to minimize crop losses and promote sustainable agricultural practices. Furthermore, beyond its immediate implications for banana cultivation, this research contributes valuable insights to the broader field of agricultural technology, showcasing the transformative potential of machine learning in addressing real-world agricultural challenges.

By the study of Arman et al. (2023) published the BananaLSD dataset, a benchmark collection of banana leaf images designed for the classification of banana leaf diseases using machine learning. Images were gathered from banana farms in Bangladesh, covering disease categories including Sigatoka, Cordana, and

Pestalotiopsis. The dataset was specifically developed to support the training of computer vision and machine learning models for plant disease detection. This study is directly relevant to the present research as it involves the same target diseases and highlights the critical role of well-curated image datasets in training reliable classification models. The present study similarly relies on a collected dataset of banana leaf images from local plantations as the foundation for model training and validation.

The study by A. Kalaivani et al. (2024) developed a mobile application that uses a Convolutional Neural Network (CNN) to detect and classify banana leaf spot diseases. The research highlights the importance of early disease detection in improving crop yield and assisting not only farmers but also beginners and entrepreneurs in agriculture. One notable feature of the system is the inclusion of voice assistance and remedy recommendations, making it more accessible to users with limited agricultural knowledge. The diseases using deep learning, achieving 98.7% accuracy with MobileNetV2. The application was deployed on Android devices running version 4.1 or later and included disease information and symptom galleries accessible without internet access. This study is particularly relevant to the present research as it demonstrates the full development cycle of an offline Android application for crop disease detection, from model training to mobile deployment, mirroring the methodology employed in the current study. The success of offline deployment in potato disease detection supports the feasibility of applying the same

approach to banana leaf disease detection in rural, connectivity-limited environments.

The study by Chaudhari and Patil (2023), proposed a system for banana leaf disease detection and classification using a novel segmentation technique and an ensemble machine learning approach. The researchers applied image processing steps such as pre-processing, filtering, and contrast enhancement, followed by image segmentation using a genetic algorithm. Feature extraction was performed using Local Binary Pattern (LBP), and multiple classifiers were combined through an ensemble method, achieving an accuracy of over 92%. This study is related to the present research as the current system employs GLCM for texture feature extraction as part of its image processing pipeline. The findings of Ossai and Wickramasinghe affirm the effectiveness of GLCM as a feature extraction method for automated plant disease detection, supporting its use in the proposed banana leaf disease classification system.

Javidan et al. (2023), as cited in the *Frontiers in Plant Science* review, employed image processing algorithms and multi-class support vector machines for diagnosing and classifying grapevine leaf diseases, achieving up to 98.97% accuracy when combined with PCA and GLCM feature selection. The integration of GLCM-derived texture features with SVM classification demonstrated strong performance in distinguishing between multiple disease classes. This study is related to the present research as it validates the combination of GLCM feature extraction and SVM classification and the same pipeline used in the proposed system. The high

accuracy reported by Javidan et al. supports the suitability of this approach for multi-class plant disease detection.

The review by R and A (2025), examined various AI-powered techniques for banana leaf disease diagnosis, including machine learning, deep learning, and transfer learning approaches. The study highlighted the importance of early and accurate disease detection in maintaining banana crop productivity and emphasized the use of lightweight deep learning models for deployment in resource-limited environments. It also discussed common diseases such as Fusarium wilt, Black Sigatoka, Yellow Sigatoka, and Anthracnose, along with challenges in dataset quality and class imbalance.

Guam et al. (2021) introduced the Smarter Pest Identification Technology (SPIDTECH), an Android application designed for identifying pests and diseases in crops including banana. The system enables remote monitoring and digital diagnosis, emphasizing the importance of integrating mobile applications into agricultural management systems. This study supports the idea that mobile platforms can enhance accessibility and real-time decision-making for farmers.

Chong et al. (2025) presented an automated approach for detecting and classifying banana leaf diseases using deep learning techniques through SqueezeNet, AlexNet, and MobileNet architectures. Banana plants are highly susceptible to diseases such as Black Sigatoka, Yellow Sigatoka, and Fusarium Wilt, which significantly reduce crop yield and quality. Traditional methods of disease detection rely on manual inspection by agricultural experts, which can be time-consuming, subjective, and prone to human error. To address these limitations, the researchers

explored the use of convolutional neural networks (CNNs) for accurate, efficient, and rapid disease identification.

Sahu and Pandey (2023) developed a novel hybrid Random Forest Multiclass SVM (HRF-MCSVM) to detect plant leaf diseases. Prior to image analysis, the images were segmented and preprocessed using spatial fuzzy c-means to ensure accurate calculations. They utilized a dataset from the Plant Village dataset, which contains images of healthy and diseased leaves. The hybrid Random Forest Multiclass SVM (HRF-MCSVM) developed by Sahu and Pandey (2023) demonstrated good potential for effective detection and discovery of plant leaf diseases using combinations of image processing, feature extraction, and machine learning classification. The system achieved measurable performance based on metrics such as F-measure, which indicates a level of success in classifying healthy and diseased leaves from the Plant Village dataset. The system's effectiveness was then evaluated using performance indicators such as F-measure, accuracy, sensitivity, recall, and specificity values. The research describes the approach, but lacks in-depth explanations of how these techniques were implemented. Sahu and Pandey's (2023) research shows a promising approach for plant leaf disease detection using hybrid machine learning; however, it lacks detailed feature implementation. Future studies should clarify feature usage and SVM kernel choices, and validate the model on diverse datasets. For instance, it would be beneficial to understand how specifically the identified texture and color features were utilized. And while the provide details regarding the techniques used, including methods for extracting color and texture features, and the types of SVM kernels.

Xiojun Jin et al. (2021) addressed the difficulty of differentiating weeds from crops in complex agricultural environments. Their model focuses on identifying only the target vegetables, avoiding the challenge of handling various weed species. The approach proposed in this research for the identification of weeds is composed of two stages. The first stage consists of the state-of-art CenterNet algorithm for detecting bok choy in this study. Images of bok choy are collected and used as input data for training the neural network. The trained neural network is used for detecting bokchoy and drawing bounding boxes around them, which generates bounding box coordinates and associate class probabilities as the output. In the second stage, a color index-based segmentation is performed on vegetation outside the bounding boxes using color information, returning visual classification regarding the presence of weeds in the image. The thresholding value of the confidence scores, 1580 ground truths bounding boxes from the test set were used to evaluate the model Five. threshold values (0.5, 0.6, 0.7, 0.8 and 0.9) were used and tested. 0.6 was found as the optimal value as it yielded a precision of 95.6% and a recall of 95%. However, the study achieved excellent performance metrics like high precision and recall, the results may be limited to specific types of vegetables and weed species present in the dataset.

Cjackulins & S. Murugavalli (2022) noted that agriculture plays a significant role in India due to population growth and increasing food demands, necessitating improved crop yields. One major factor affecting low crop yields is plant diseases caused by bacteria, fungi, and viruses, which can be managed using plant disease detection techniques. Machine learning methods are widely used for disease

identification as they leverage data-driven approaches to enhance detection accuracy. This paper presents a comprehensive review of various AI-based machine learning and deep learning techniques applied in plant disease detection. The study compares machine learning and deep learning approaches, demonstrating the effectiveness of deep learning models in achieving better detection accuracy. To mitigate crop losses, Machine learning techniques can be applied to analyze leaf images for disease detection. This study emphasizes the growing role of AI-driven approaches in plant disease identification, showcasing the advantages of machine learning in improving detection accuracy and aiding agricultural productivity.

Mikhail A. Genaev et al. (2021) investigated the challenge of accurately and efficiently diagnosing fungal diseases in wheat crops, noting that such diseases can significantly reduce crop yields. The paper proposed a method for recognizing plant diseases on images obtained in field conditions based on deep machine learning. The dataset also included healthy plants and cases of multiple diseases. The image dataset represents a specially formed and labeled sample of 2414 images, which is freely available. The EfficientNet-based model achieved an impressive accuracy of 94.2%, demonstrating its capability to identify diseases even under challenging conditions. The study also implemented the recognition method as a bot on the Telegram platform, allowing for real-time disease assessment in field conditions. However, the study demonstrates high accuracy and practical implementation via a Telegram bot, it lacks a detailed exploration of its generalizability across varied environmental conditions and larger, diverse datasets critical for robust real-world applicability.

The research conducted by Jammer Kotwal (2023), For plenty of years, researchers have researched how to tackle the common disease effects among humans, animals, and plants. There are still many gaps remaining to identify and explore. According to this study there have been many researchers using Deep Learning (DL) and Transfer Learning (TL) technologies to detect agricultural diseases based on Machine Learning (ML) algorithms that were developed with the development of Artificial Intelligence (AI) technology. Many DL architectures are carried out together with numerous diverse visualization strategies to perceive and label the features of plant diseases. Our take a look at additionally makes a specialty of how ML strategies had been moved from conventional ML to DL and additionally numerous overall performance metrics (F1-score, sensitivity, accuracy, etc.) are used for the assessment of the architecture/strategies. Plant disease computerization in agriculture areas an important for every country, as the population rate increases the demand for food supply also increases. Today, the significant adaptation of modern techniques and tools increases the accuracy of detection of plant disease. Identifying plant diseases in an early stage can reduce their spread. Early identification is a beginning stage to fight against disease spreading. However, the study lacks a structured analysis of the existing gaps in plant disease detection and does not provide concrete solutions to overcome the challenges associated with real-world implementation.

Ananda S. Paymode & Vandana B. Malode (2021) established that the use of artificial intelligence in agriculture has become crucial for improving productivity. Controlling diseased leaves during the growing stages of crops is essential, and early

disease detection, classification, and analysis aid in agricultural progress. The detection and classification of diseases in crops such as tomatoes and grapes are the primary focus of this study. The study utilizes Convolutional Neural Network (CNN) methods for detecting Multi-Crop Leaf Disease (MCLD). Machine learning-based models extract features from images to classify diseased and healthy leaves, with the CNN-based Visual Geometry Group (VGG) model enhancing performance. A dataset of crop leaf images was used for training and testing, with performance metrics such as accuracy, sensitivity, specificity, precision, recall, and F1-score monitored. The proposed model aims to improve classification accuracy, achieving 98.40% for grapes and 95.71% for tomatoes. This research contributes to increasing agricultural productivity by enhancing the efficiency of disease detection through machine learning-based approaches. While this study presents promising results, future research could explore real-time deployment, robustness against environmental variations, and integration with IoT-based monitoring systems for practical agricultural applications.

Debnath Oliva & Himadri Nath Saha (2022) reported that rice farmers face significant financial setbacks, with diseases like Brown Spot reducing yields by 15-20% annually. This paper presents a novel method of detecting fungal brown spots occurring in rice paddy using convolutional neural networks in integration with IoT. The novelty of this work also includes that no predefined dataset has been used here. The model has been trained and tested based on real time data which is far more challenging. OpenCV are implemented using Android Studio and Scikit-learn frameworks. These models are trained on manually collected datasets to identify

diseases like Brown Spot with high accuracy. The study evaluates the model's accuracy, achieving a reported 97.70% accuracy in disease detection. While the study mentions the use of IoT and deep learning in real-time scenarios, there seems to be limited information on field testing. This raises concerns about how the model and IoT systems perform under real-world conditions with variable factors like lighting, weather, and network latency.

Damandeep Kaur (2023) established that conventional methods for detection can be time-consuming, labor-intensive thus, developing an optimal method with automation is thus imperative. The aim of this study is to develop an accurate and efficient approach for diagnosing disease symptoms, and an optimal method to solve the problem in an efficient and cost-effective manner. Quantum-based classifiers also play an essential role in detecting the leaf profile features in recognizing leaves using a computer vision system that can be used with a quantum computer. Enhanced double quantum image encryption (EDQRCI) enables automation to produce and create quantum pictures that transform color pixels into well-defined shapes. The proposed multi-layer convolutional neural network model is a good classification model that classifies mango leaves infected with anthracnose fungal disease accurately. However, the research lacks clarity and coherence in explaining, how quantum computing is effectively utilized in the classification process.

Harakannanavar et al. (2022) noted that long-term validation under varying environmental conditions is necessary to fully assess the reliability of plant disease detection systems. Seasonal variations, such as increased fungal infections during rainy periods, may affect system performance over time. The offline functionality

of the application, although essential for accessibility, also presents limitations. According to the Office-in-Charge of Agriculture, many farming areas in Rosario have limited or unstable internet connectivity, making offline operation a necessary feature. However, this design prevents the system from performing real-time updates, accessing cloud-based databases, and continuously improving its accuracy through new data.

Acevedo-Sánchez et al. (2025) established that machine learning models are highly dependent on the diversity and quality of the dataset, and limited datasets may reduce the model's ability to generalize to new or unseen data. Variations in lighting conditions, image resolution, leaf orientation, and background noise may further affect the accuracy of the system when used under actual field conditions. Furthermore, the application focuses exclusively on leaf symptoms. Diseases affecting the roots, pseudo stems, or internal vascular system, which may not show leaf symptoms in the very early stages, are outside the diagnostic capability of this system.

The United States Department of Agriculture (USDA, 2022) noted that some major banana diseases, such as Fusarium wilt, originate in the soil and vascular system before visible symptoms appear on leaves. Similarly, certain viral and fungal infections may progress internally before external symptoms become evident. Therefore, the system may not detect diseases at their earliest stage if leaf symptoms have not yet developed, which limits its diagnostic capability to visible leaf conditions only.

Zamani et al. (2022) described how precision agriculture's automatic leaf disease detection system employs image acquisition, image processing, image segmentation, feature extraction, and machine learning techniques. An automated disease detection system offers the farmer with a fast and accurate diagnosis of the plant disease. Automation of plant leaf disease detection system is essential for accelerating crop diagnosis. Using machine learning and image processing, this paper describes a framework for detecting leaf illness. An image of a leaf can be used as an input for this framework. To begin, leaf photographs are preprocessed in order to remove noise from their images. The mean filter is used to filter out background noise. Histogram equalization is used to enhance the quality of the image. The division of a single image into multiple portions or segments is referred to as segmentation in photography. It assists in establishing the boundaries of the image. Segmenting the image is accomplished using the K-Means approach. Feature extraction is carried by using the principal component analysis. Following that, images are categorized using techniques such as RBF-SVM, SVM, random forest, and ID3.

Kumaran et al. (2024) explored machine learning approaches for Early Detection and Classification of Cervical Cancer Using Image Processing Traditional machine learning (ML) techniques have limitations that make it difficult for existing algorithms to diagnose cervical cancer. These limitations include lower accuracy and an inability to handle complicated variations in histopathological images. These systems usually rely on manual evaluations, decision trees, logistic regression, and basic neural networks; these methods are not well suited to handling the nuances of

precancerous lesions and picture quality fluctuations. By combining innovative techniques, the proposed hybrid system solves these problems. Specifically, it integrates Convolutional Neural Networks (CNNs) for hierarchical feature extraction, Vision Transformers (ViTs) for contextual information and ensemble methods that combine Gradient Boosting Machines (GBMs) and Support Vector Machines (SVMs) for robust classification. Further improving model robustness is self-supervised learning. The methodology outperforms the existing systems in terms of accuracy (97. 0%), precision (94. 5%), recall (95. 2%), and F1-score (94. 8%), indicating improved performance and fewer false positives and negatives.

Keerthishree et al. (2025) aimed to enhance the predictive accuracy of eye cancer by integrating advanced image preprocessing techniques. A comprehensive dataset of digital histopathological images is utilized, undergoing a series of preprocessing steps to improve data quality and consistency. These steps include RGB-to-grayscale conversion, thresholding, image sharpening, segmentation, noise reduction, contrast stretching, histogram modification, and feature extraction. These techniques help refine image clarity, enhance critical features, and reduce artifacts, ensuring optimal input for machine learning models. The main goal of this study is to use histopathological image processing to create an accurate and effective model for the early detection of leaf. To avoid serious leaves issues including black sigatoka, early detection is crucial. This initiative is a big step toward bettering the diagnostic tool, which could save lives and improve the standard of treatment for those diseased.

Park et al. (2021) investigated edge detection challenges in computer vision, noting that despite ongoing advances in the field, reliable edge detection remains difficult. It comes from the limitations of the complementary metal oxide semiconductor (CMOS) Image sensor used to collect the image data, and then image signal processor (ISP) is additionally required to understand the information received from each pixel and performs certain processing operations for edge detection. Even with/without ISP, as an output of hardware (camera, ISP), the original image is too raw to proceed edge detection image, because it can include extreme brightness and contrast, which is the key factor of image for edge detection. To reduce the onerousness, we propose a pre-processing method to obtain optimized brightness and contrast for improved edge detection. In the pre-processing, we extract meaningful features from image information and perform machine learning such as Random Forest and support vector machine (SVM) to obtain enhanced model by adjusting brightness and contrast. The comparison results of F1 score on edgy detection image of non-treated, pre-processed and pre-processed with machine learned are shown. In addition, if we go through the pre-processing method that we proposed, it is possible to more clearly and easily determine the object required when performing auto white balance (AWB) or auto exposure (AE) in the ISP. It helps to perform faster and more efficiently through the proactive ISP.

REVIEW OF RELATED STUDIES

SYNTHESIS

RESEARCHERS	TITLE	METHODS	DATA SET	PERFORMANCE	DIFFERENCES
Selvaraj et al. (2021)	Banana disease detection using UAV and multi-spectral satellite imagery for pixel-based classification and disease localization of banana plantations.	SVM, Random Forest, image resizing, preprocessing and pixel-based classification.	The datasets consists of UAV-RGB and multi-spectral satellite images of banana plantations in field landscapes.	99.4% healthy banana, 92.8% BBTD, 93.3% BXW, 90.8% individual banana plants.	Uses expensive aerial drone and satellite imagery rather smartphone camera capture; not designed for farmer-operated mobile devices in remote areas; no offline application developed.

Sahana S A, et al (2025)	Interpretable Machine Learning framework using K-means, GLCM texture extraction, HSV color histogram s, and SVM classificat ion.	K-means, GLCM, HSV color histograms and SVM.	The dataset consists of general leaf image covering multiple plant species.	High accuracy maintained in low-resource computing environments without heavy computational demands.	Not specific; covers general diseases only; no offline mobile application was developed or tested in actual field conditions.	banana-leaf
Sathya et al. (2024)	Machine learning-based system for	Random Forest, Support Vector Classificati	The datasets consists of healthy and	The model accurate classification of healthy versus diseased	No deployment; does not conduct a direct	mobile

	banana	on	and	diseased	banana	leaves	controlled
	leaf	image		banana	demonstrated.		comparison of
	disease	dataset		leaves			SVM and RF
	detection	preprocessi		image.			under identical
	and	ng.					experimental
	classificat						conditions.
	ion using						
	Random						
	Forest						
	and						
	Support						
	Vector						
	Classifica						
	tion to						
	distinguis						
	h healthy						
	from						
	diseased						
	leaves.						
Ahmed et al. (2023)	Multi- stage automate	K-means clustering for	The dataset consists of general	Demonstrated feasibility removing	Multi-crop focus with no banana-specific		

	d	leaf segmentati	image	manual errors	application; no
	disease	on, GLCM	dataset	from the	mobile
	detection	texture	from	diagnostic	application
	using K-	feature	multiple	process	developed for
	means	extraction,	crop types.	through	farmer field
	segmentat	SVM,		automated	use.
	ion,	KNN, and		classification.	
	GLCM	CNN.			
	feature				
	extraction				
	, and				
	multiple				
	machine				
	learning				
	classifiers				
	to reduce				
	manual				
	diagnosti				
	c error.				
Chaudar	Fruit and	SVM,	The dataset	Reliable	and Broader fruit
i et al.	lead	Random	consists of	efficient	and multi-crop
(2024)	disease	Forest,	images of	classification	focus; not

classificat image fruits and of multiple exclusively
 ion using resizing, leaves diseases banana-
 image color including including the specific; no
 processin normalizati banana- three diseases offline mobile
 g and on, related targeted in the application
 machine segmentati disease present study. developed for
 learning on, texture samples. farmer
 technique and color deployment.
 including feature
 SVM and extraction.
 Random
 Forest to
 detect
 diseases
 such as
 Black
 Sigatoka,
 Cordana
 Leaf
 Spot, and
 Pestalotio
 psis Leaf
 Blight

Praneetha and Tabassum (2025)	IoT-based precision agriculture system for banana disease monitoring using GLCM feature extraction, Random Forest classifier, Raspberry Pi 3, cloud server integration, IoT sensor data.	Specialized hill banana dataset with IoT sensor and image data.	Effective disease monitoring demonstrated for specialized banana cultivars.	Requires external IoT hardware and continuous cloud internet connectivity; not suitable for remote farming areas without stable internet access.
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Baraj et al (2024)	Numerical	Mobile-D	The dataset	Successful	Targets
	1	agile	consists of	early	different
	applicatio	methodolo	banana	identification	banana diseases
	n	gy,	disease	of targeted	from the
	developm	machine	images	banana	present study;
	ent for	learning	covering	diseases	does not
	early	models for	Moko,	through	conduct a
	identificat	banana	Bacteriosis	functional	direct
	ion of	disease	and Black	mobile	structured
	banana	classificati	Sigatoka.	platform.	comparison
	plant	on			between SVM
	diseases				and RF
	using				classifiers.
	machine				
	learning				
	models				
	and the				
	Mobile-D				
	agile				
	developm				
	ent				
	methodol				

Zamani et al. (2022)	Comprehensive leaf disease detection combining K-means segmentation, PCA feature extraction, and multiple machine learning classifiers	K-means segmentation, Principal Component Analysis (PCA), RBF-SVM, PCA SVM, Random Forest and ID3.	The dataset consists of general leaf image dataset from multiple crop types	Efficient and accurate automated disease diagnosis demonstrated across multiple classifiers.	and Multi-crop focus with no banana-specific application; no deployment or offline functionality developed.
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including

SVM and

random

Forest.

Jianqing et al. (2022)	Banana Leaf disease detection using YUV color space conversion and Otsu Segmentation to isolate disease regions from	YUV color space conversion, Otsu segmentation on minimum Euclidean distance classification based on color mean and variance.	Field-collected banana leaf images from actual plantation environments.	91.7% recognition rate of Leaf Spot, 90% recognition rate for Sigatoka.	Does not use GLCM texture features or advanced classifiers such as SVM or RF; limited to two diseases only; no mobile application developed.
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complex
plantation
backgrou
nds for
field-
collected
images.

Gueco et al. (2023)	Field survey document ing banana disease prevalenc e and farming practices among smallhold er banana growers in Luzon	Field survey, agricultural documenta tion, disease observatio n among smallholde r farmers.	Local farm survey data from smallholde r banana growers across Luzon and Visayas	Identified and documented the most economically damaging banana diseases affecting Philippine banana farms.	Contextual and literature study only, no machine learning, image processing, or mobile application component involved.
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and

Visayas

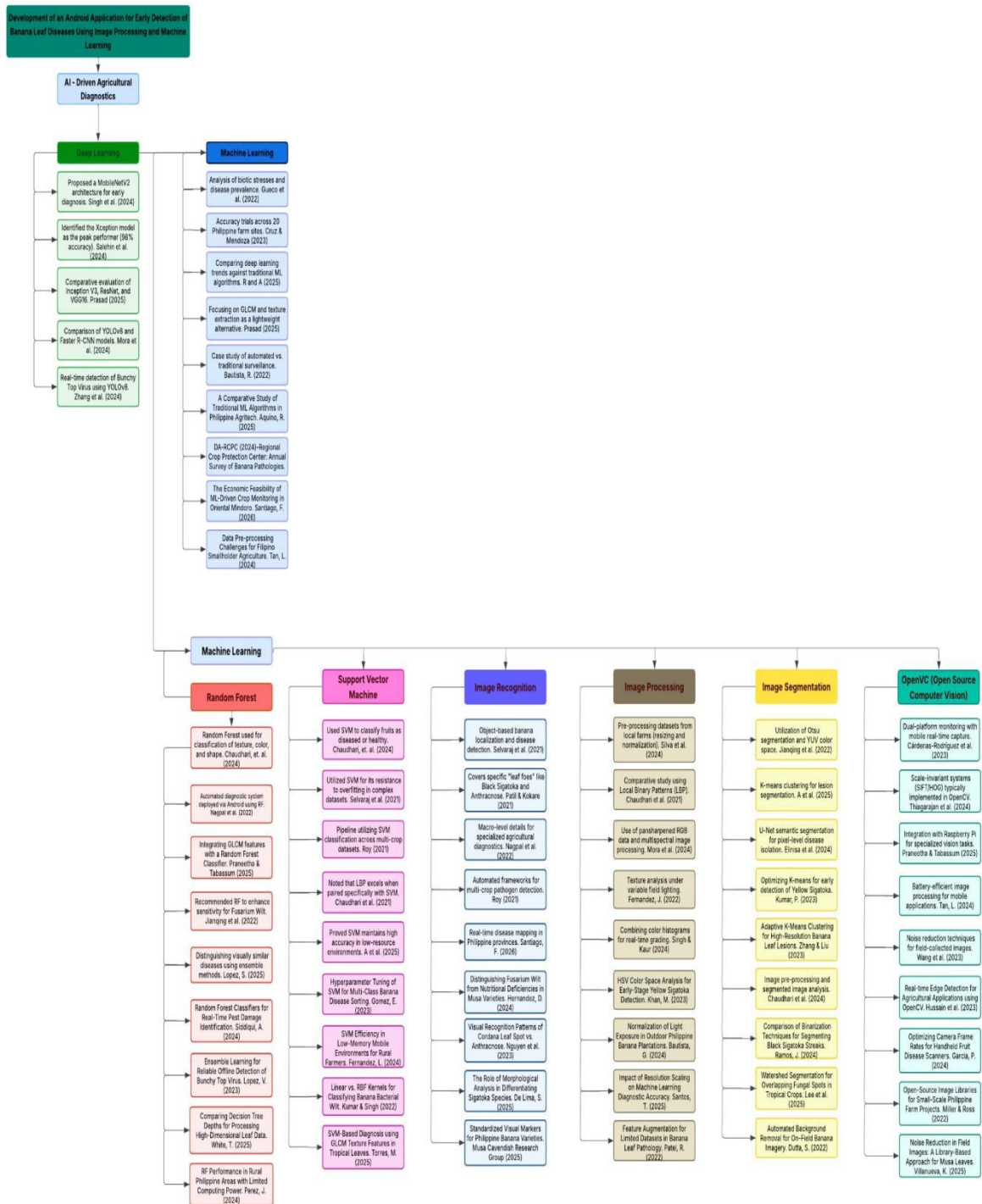
in the

Philippin

es

context.

LITERATURE DIAGRAM



CONCEPTUAL FRAMEWORK

In this Conceptual framework, we use input-process-output method as shown in figure. This framework will show the process of our system.

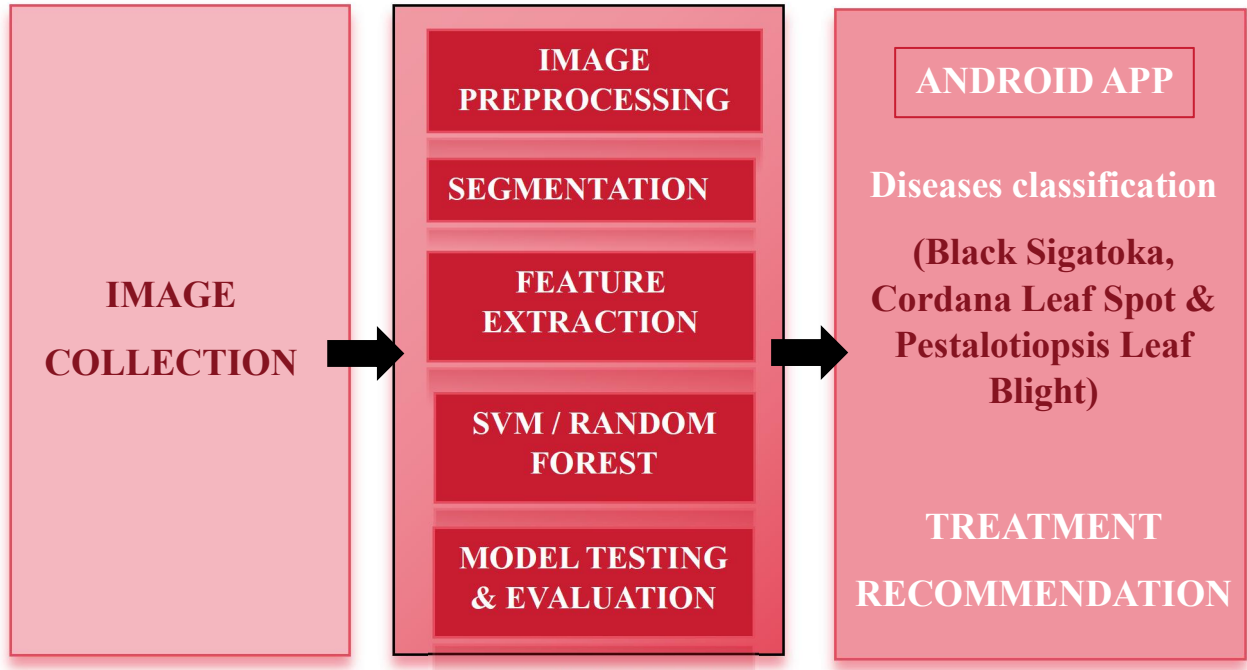


Figure 11. *Conceptual Framework*

The input phase starts with the collection of banana leaf images using Android camera and existing datasets. These images include both healthy and diseased leaves affected by Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight, which used for training and testing the system. Through the process phase, the images undergo preprocessing such as resizing, noise removal, and enhancement. Segmentation is then applied using K-means clustering or Otsu's Thresholding to isolate diseased areas. Feature extraction follows using GLCM, and color histogram techniques. The extracted features are used to train and test machine learning models, specifically Support Vector Machine (SVM) and Random Forest (RF), for disease classification and Performance evaluation. Lastly, the output phase

is the deployment of the trained model in an Android application that detects and classifies banana leaf diseases in early stage. The system identifies whether the leaf is healthy or diseased and specifies the disease type, along with providing recommendations to support farmers in early disease management.

DEFINITION OF TERMS

In order to know all the conceptual terms used in this study the following are needed to be reviewed.

OPENCV (OPEN-SOURCE COMPUTER VISION) - is a powerful, open-source toolkit designed for handling real-time computer vision challenges and advanced image processing tasks. It enables automated systems to interpret images and videos effectively.

IMAGE SEGMENTATION - is the process of dividing an image into different regions or segments to isolate specific objects or areas of interest for easier analysis.

IMAGE PROCESSING - is the manipulation and analysis of images using computer algorithms to enhance quality, extract useful information, and prepare them for further tasks such as classification or recognition.

IMAGE RECOGNITION - is a process in computer vision that enables a system to identify and classify objects, patterns, or features within an image.

SUPPORT VECTOR MACHINE (SVM) - is a machine learning algorithm that classifies data by finding the optimal boundary (hyperplane) that separates different classes.

RANDOM FOREST - is a machine learning algorithm that uses multiple decision trees to classify data and improve accuracy by combining their results.

MACHINE LEARNING - refers to a branch of artificial intelligence that enables systems to automatically learn patterns from data and make predictions or decisions without being explicitly programmed.

BLACK SIGATOKA - caused by *Mycosphaerella fijiensis* is a fungal disease of banana plants that causes dark streaks on leaves, leading to reduced photosynthesis, premature leaf death, and lower crop yield.

CORDANA LEAF SPOT - the fungus caused by *Neocordana musae*. is a fungal disease of banana plants that causes oval to irregular brown or gray leaf spots, which can reduce leaf function and affect plant growth and productivity.

PESTALOTIOPSIS LEAF BLIGHT - is a fungal disease of banana plants that causes leaf spots and blight, often starting as small dark lesions that expand and lead to leaf drying, reduced photosynthesis, and decreased plant health.

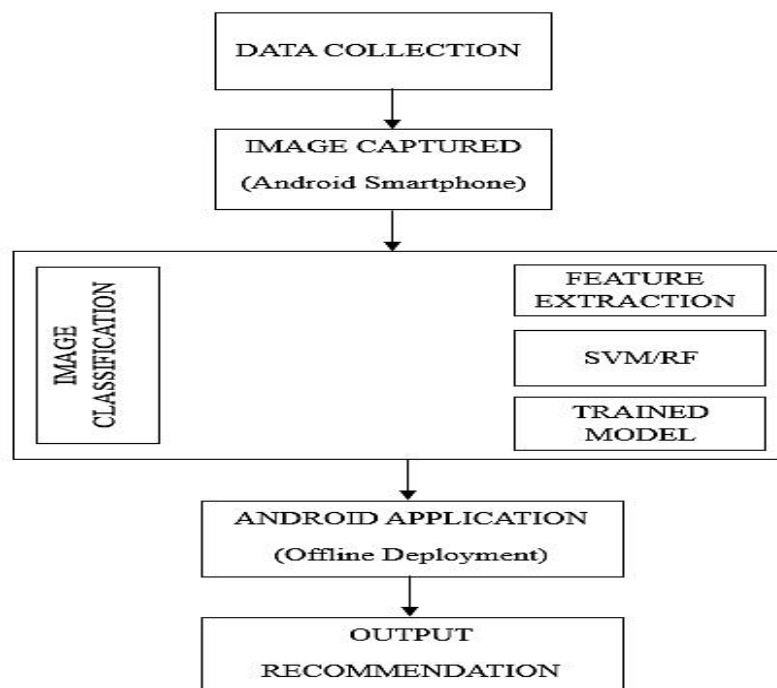
CHAPTER III

RESEARCH METHOD AND PROCEDURE

This chapter represents the development of an Android application designed for the early detection and analysis of diseases in banana leaves to help solve local agricultural problems. It follows structured process involving image acquisition, preprocessing, segmentation, and classification to ensure the system is fast and accurate without needing the internet. Modern tools and methods, such as K-means clustering and machine learning, are used to identify diseases from leaf images. This easy-to-use approach helps all farmers, including elderly, detect problems early to protect their crops and improve their harvest.

RESEARCH METHOD

System Diagram



DATA COLLECTION

The process of data collection for our study of banana leaf diseases commenced with a systematic and precise planning stage, wherein the researchers established clear research questions and objectives focused on early disease detection. On the basis of this, the researchers conducted suitable methodologies, such as the systematic observation of the morphological features of banana leaves. The researchers employed an orderly sampling approach, selecting leaf specimens from various plantations and different positions on the pseudo stem to ensure a diverse datasets. We also took into consideration the data provided by the Office of Agriculture in Rosario, Batangas, to identify the most prevalent threats to local crops. The leaf samples were handled with care and documented in their natural environment. The researchers used an Android smartphone to take high-resolution pictures of these samples, ensuring the images captured both healthy tissues and subtle disease patterns. All information collected, including direct field observations, digital photographs, and local agricultural data, was carefully noted and documented. This organized repository serves to make the subsequent image processing and machine learning classification easier, forming the solid foundation of evidence for our research into protecting local banana plantations.

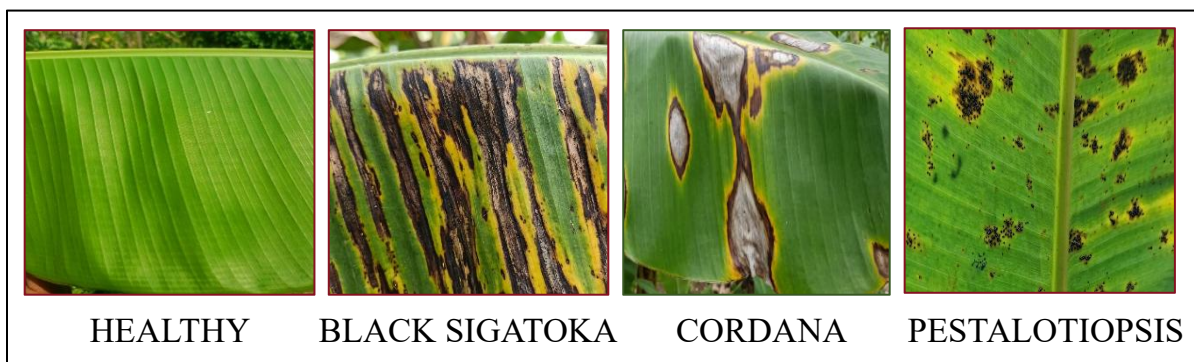
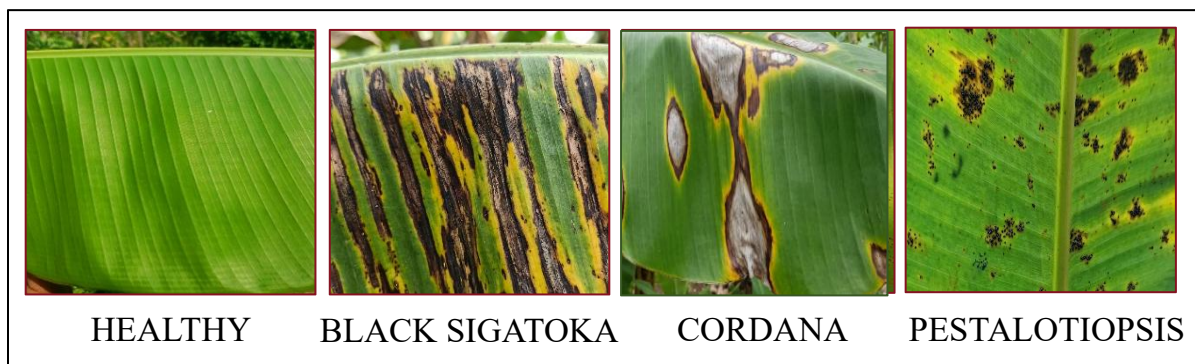


Figure 13. *Data Collection*

DATASET COMPILATION

Datasets compilation involves collecting high-resolution images of banana leaves directly from local farms in Rosario, Batangas using Android devices. The collection captures a diverse range of conditions specifically targeting those infected leaves and the healthy ones. To ensure model resilience, shadows, and complex backgrounds. The raw images undergo a rigorous preprocessing pipeline: they are manually annotated by agricultural experts, resized for computational efficiency, and normalized to ensure consistency. To improve classification performance of the SVM or Random Forest models, data augmentation techniques (such as rotation, scaling, and flipping) are applied to the training set. The final datasets is partitioned into training, validation, and testing sets. Unlike many existing studies that utilize laboratory-controlled-controlled datasets (like Plant Village), this research prioritizes real-world variability. By focusing on images captured via smartphone in the field, the dataset is specifically tailored for low-latency, on device inference. This ensures that the resulting Android application remains robust and reliable for Filipino farmers performing offline, instantaneous diagnoses in rural areas.



TRAINING

The model is developed using a specialized database of banana leaf images featuring healthy samples and those infected. To ensure consistency images are preprocessed using OpenCV through resizing and normalization, while data augmentation is applied to help the model recognize diseases in various field conditions. Unlike standard deep learning, the system utilizes K-means clustering and Otsu's thresholding to isolate infected tissues from the healthy leaf. From these segments, the system extracts critical identifiers such as Color Moments and GLCM texture features. These features are then used to train a Support Vector Machine (SVM) or Random Forest classifier. The dataset is divided into training, validation, and testing sets to optimize diagnostic accuracy and prevent overfitting. The resulting lightweight model is integrated into an Android application, allowing it to perform high-speed, offline classification. This iterative process ensures that the tool provides Filipino farmers with instant, reliable results without the need of an internet connection.

IMAGE PROCESSING

Image processing is essential for accurately identifying banana leaf diseases, by examining the visual symptoms like spots, lesions or discoloration. By utilizing OpenCV, the system performs real-time segmentation to isolate infected tissues from healthy part of the leaf. This approach allows the system to focus exclusively on the

diseased sections of the leaf, ensuring that the extracted color and texture data are accurate and free from the “noise” of healthy tissue. By processing images directly on a smartphone, the technology allows for instant detection and response in remote fields without internet access. This technology eventually aids sustainable farming by minimizing dependence on human inspection and chemical treatment.

IMAGE PREPROCESSING

To ensure consistency in the input data, this step includes operations such as resizing, normalization, and noise reduction. This method helps the SVM or Random Forest models detect banana leaf diseases more accurately by standardizing the images before feature extraction. It also reduces variations caused by uneven lighting, background clutter, and different camera angles, making the offline detection process more reliable. By refining the visual content of the captured smartphone images, the system can better isolate diseases. These refined images are optimized to improve quality, ensuring that the segmentation and classification stages can identify fungal and viral infections with high precision.

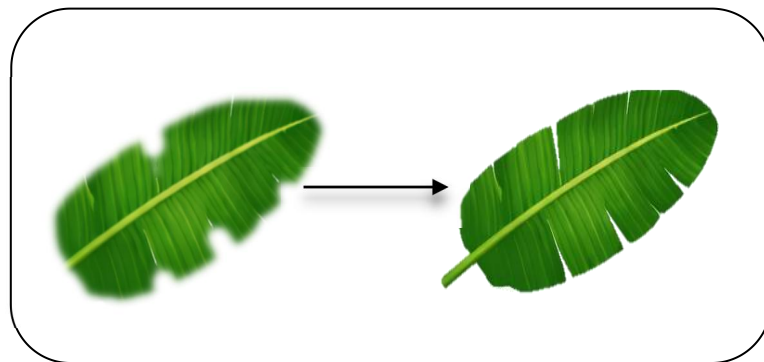


Figure 15. *Image Preprocessing*

IMAGE CLASSIFICATION

Image classification is crucial to help the system automatically detect and classify banana leaves depending on their health status. Through image classification, the application is able to scan images taken of banana leaves and identify whether they are infected by certain diseases like Black Sigatoka or Cordana leaf Spot. This procedure does away with the manual inspection by professionals, and therefore, early detection becomes quicker, more convenient, and scalable to farmers. Consequently, image classification is the central activity that powers the entire system—initiating treatment suggestions, producing reports, and eventually assisting farmers in making the right decisions to safeguard and maintain their banana plants better.

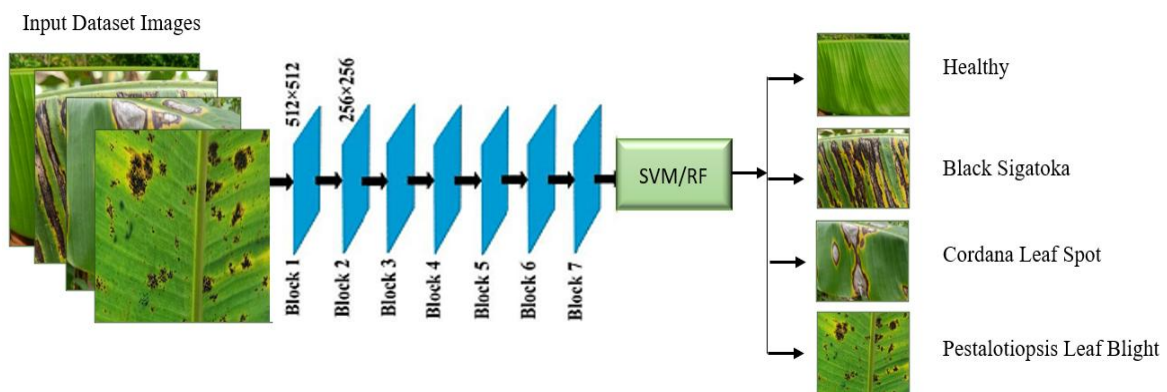


Figure 16. *Image Classification*

MACHINE LEARNING

Machine Learning is a subset of artificial intelligence (AI) that enables systems to learn from data, identify patterns, and make decisions or predictions without explicit programming. By feeding large datasets, such as images of healthy and diseased banana leaves, models improve their diagnostic accuracy through experience.

In this study, machine learning is applied through the Support Vector Machine (SVM) or Random Forest (RF) algorithms. Instead of relying on a constant internet connection, the system uses these pre-trained models to analyze the specific texture and color features of a leaf on-site. This allows the application to categorize infections like Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight with high precision. By learning from a diverse database of agricultural samples, the system provides farmers with a reliable, automated tool that replaces the need for manual expert intervention.

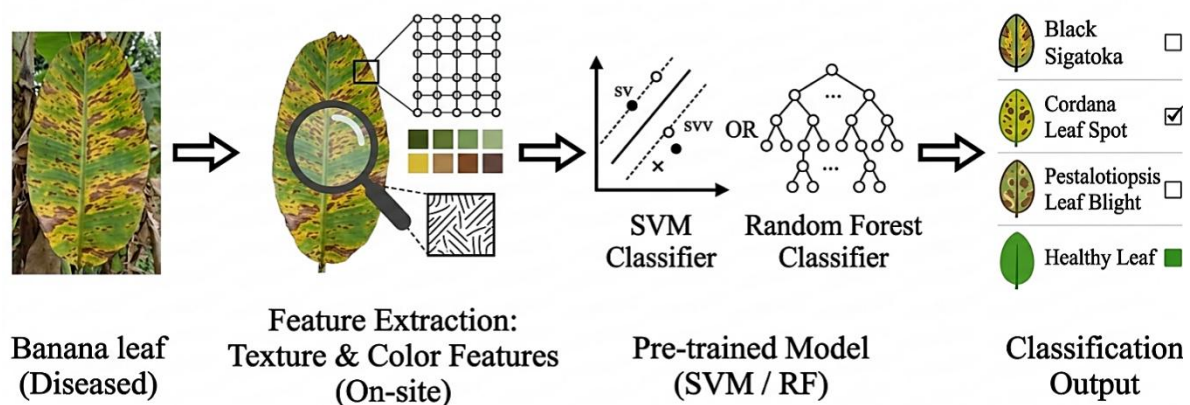


Figure 17. *Machine Learning*

LEAF DISEASE CLASSIFICATION

The images of banana leaves undergo labeling and are subsequently used to train machine learning model based on Support Vector Machine (SVM) or Random Forest classifier. These models are highly effective for classification tasks in resource-constrained environments. During the training process, extracted features, such as color and texture data, are fed to the model so that it can learn to identify specific diseases patterns. Once trained, the model is tested using a separate dataset to ensure its diagnostic accuracy and reliability. Validated, the trained model is then implemented in an Android application that allows users, especially banana farmers, to scan leaf images in early stage. The app processes the image, classifies the disease, and supplies corresponding treatment solutions, thus presenting a useful and accessible solution to early detection and management of leaf diseases in banana farming.

TRAINED MODEL

A trained model is essential to identify leaf diseases in banana through image processing and machine learning since it will allow for accurate detection of patterns and symptoms of disease. By learning from annotated data, the model can classify healthy and diseased leaves despite slight differences between them. Farmers can then use this to intervene early enough before infections spread, saving crops and increasing yield. Besides, trained models have the ability to automate the detection process, thereby conserving time and resources relative to manual inspection.

Generally, they support sustainable agriculture by promoting better disease management practices.

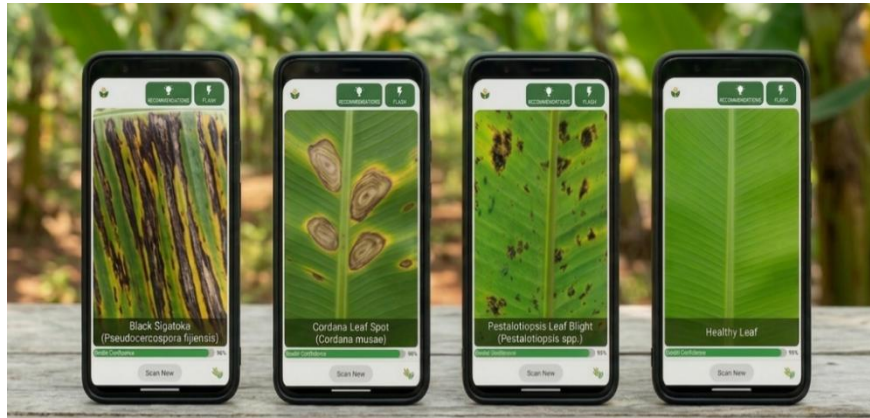


Figure 18. *Trained Model*

TESTING

The testing of banana leaf diseases using machine learning is done to determine the accuracy and viability of output of the algorithm with respect to the actual measures in the real world.

TESTING MODEL

The flowchart describes the process for identifying leaf diseases in banana. The process starts after capturing the banana leaf using an Android smartphone camera. The image is then processed to identify specific types of fungal infections, such as Black Sigatoka, Cordana leaf spot, and Pestalotiopsis leaf blight. Lastly, the identified banana leaf disease and its corresponding treatment recommendations are displayed on the smartphone. This innovative approach has huge potential for

improving agriculture and helping traditional and new generation farmers by providing a fast, offline diagnostic tool that supports timely crop management and protects local yields.

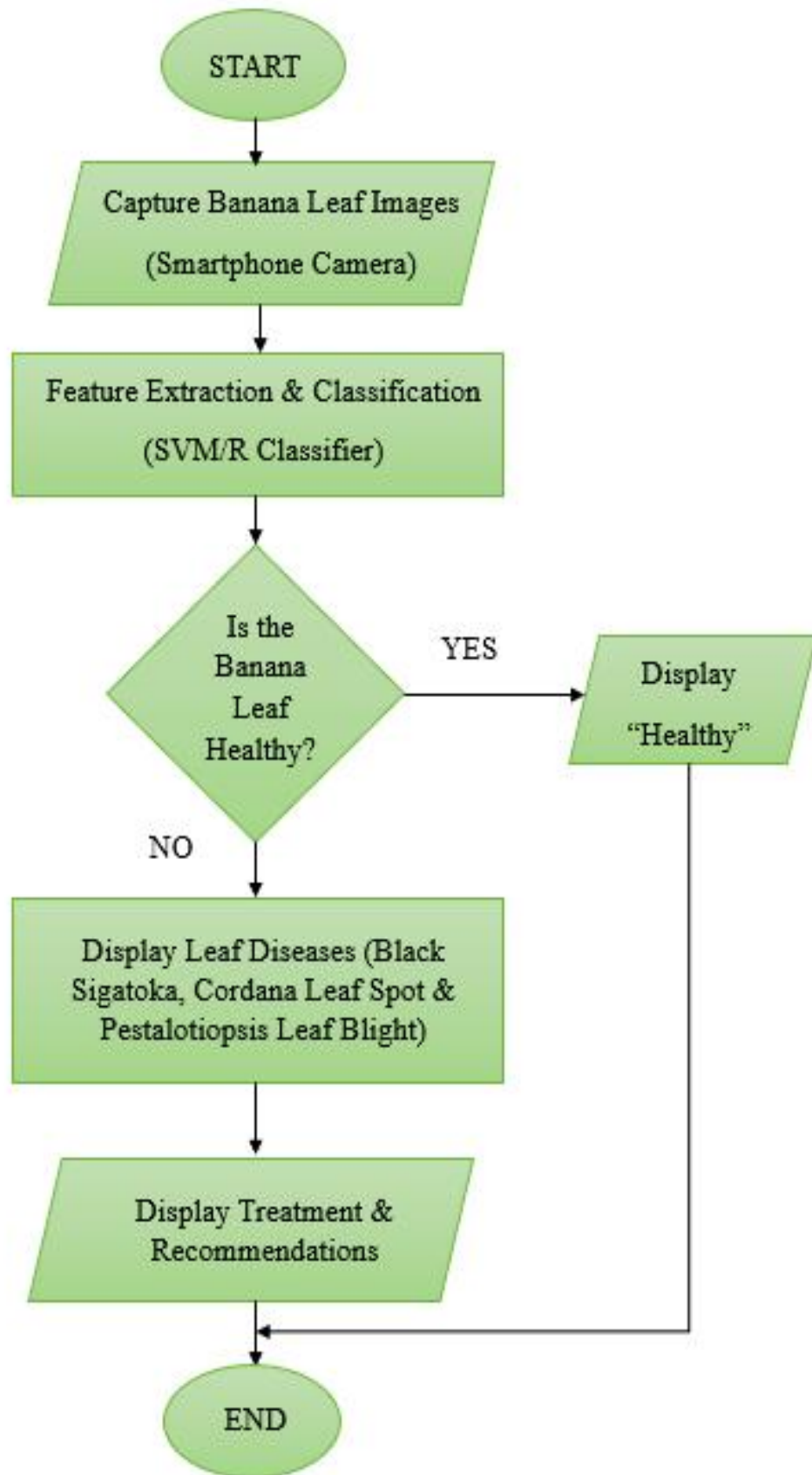


Figure 19. *Flowchart*

PROTOTYPE CONFIGURATION

HARDWARE	SPECIFICATION
DESKTOP COMPUTER	• AMD RYZEN 5 4600G (Radeon Graphics) • 16 GB RAM • SSD (For Model Training/Development)
SMARTPHONE	• Android Smartphone • At least 4 GB RAM • At least 13 megapixel (back camera)

Table 1. *Hardware*

SOFTWARE	SPECIFICATION
GOOGLE COLAB	T4 GPU (for SVM/RF model training)
ANDROID STUDIO	Latest Version ; app development IDE
LIBRARIES/TOOLS	OpenCV, Scikit-learn, Color Moments, GLCM

Table 2. *Software*

SOFTWARE DEVELOPMENT METHODOLOGY

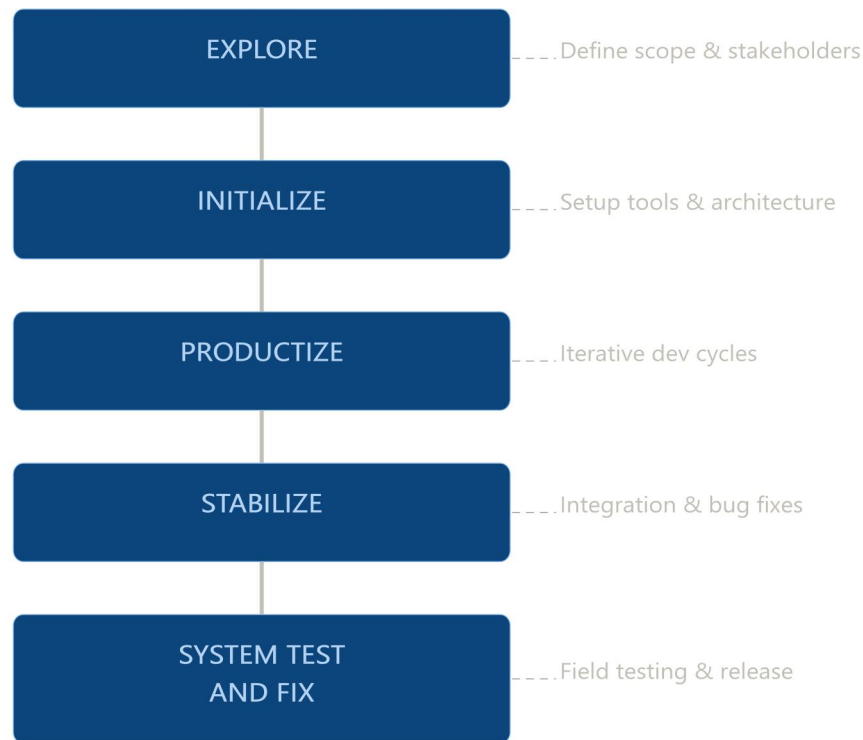


Figure 13. *The Mobile-D Agile Software Development Life Cycle*

For the development of the banana leaf diseases detection application, this study employs the Mobile-D Agile Software Development Life Cycle (SDLC). Mobile-D is an agile methodology specifically designed for mobile application development, combining practices from Extreme Programming (XP), Crystal Methodologies, and Rational Unified Process (RUP). Unlike the sequential Waterfall approach, Mobile-D is highly iterative and adaptive, allowing the development team to respond quickly to changes and continuously refine the application through short development cycles called iterations. This methodology is particularly well-suited for mobile application projects because it emphasizes rapid delivery, close

collaboration, and continuous testing, ensuring that the application meets user needs effectively and efficiently.

The development of the system follows the five core phases of Mobile-D:

1. **Explore:** This is the initial phase where the development team establishes the project foundation. The researchers define the scope of the application, identifying the three target diseases to be detected: Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight. Stakeholders, including farmers and agricultural experts, are consulted to understand their needs and constraints. The technical environment is set up, including the selection of Android Studio as the development platform and the identification of necessary libraries such as OpenCV and Scikit-learn for image processing. The project requirements and a preliminary plan for the iterative development cycles are also established during this phase.
2. **Initialize:** In this phase, the researchers prepare all the necessary resources and configurations before the actual coding begins. This includes finalizing the system architecture, setting up the development environment, and preparing the initial dataset of banana leaf images collected from local plantations. The user interface layout is drafted to ensure a simple, farmer-friendly design that allows easy camera access and clear display of diagnostic results. The machine learning models, specifically the Support Vector Machine (SVM) and Random Forest classifiers, are configured, and the image processing pipeline involving HSV color conversion, K-means clustering, and GLCM feature extraction is designed and ready for implementation.

Requirements Analysis

Following the Explore and Initialize phases, the researchers conducted a Requirements Analysis to formally define the scope, users, and specifications of the banana leaf disease detection application. This analysis ensured that all development efforts in the succeeding phases were grounded in clearly identified needs and constraints.

Problem Overview. Banana farmers, particularly those in rural areas, lack access to a fast, reliable, and offline tool for diagnosing banana leaf diseases. Existing solutions either require internet connectivity, depend on the availability of agricultural specialists, or are not designed for mobile field use. This gap motivated the development of an Android application capable of detecting Black Sigatoka, Cordana Leaf Spot, and Pestalotiopsis Leaf Blight directly on the device without requiring an internet connection.

Stakeholders and Users. The primary users of the system are banana farmers who need a simple, field-ready tool to photograph leaves and receive instant disease diagnoses. Secondary users include agricultural extension officers who may assist farmers in interpreting results. Stakeholders include the research team, agricultural institutions, and local government agricultural offices that support crop disease monitoring.

Functional Requirements. The functional requirements define the core actions the system must perform. The system shall allow users to capture or upload images of banana leaves, analyze those images to detect and classify diseases, display the

diagnosis result clearly, and provide corresponding treatment recommendations, all without requiring an internet connection. The system shall support the Android mobile platform and complete analysis within a reasonable response time.

Non-Functional Requirements. The non-functional requirements describe the quality standards the system must meet. The application shall respond quickly after an image is submitted, maintain stable performance during repeated use, and operate reliably on mid-range Android devices. The interface shall be simple and navigable without prior technical training, making it accessible to farmers with varying levels of digital literacy.

Constraints and Assumptions. The system is constrained to detecting three specific diseases and must operate in a fully offline environment. It will be deployed exclusively as an Android application, and the device must have a functional camera. It is assumed that users will have access to an Android smartphone, that submitted images will be of banana leaves taken under visible field conditions, and that the machine learning models will be fully trained and validated prior to deployment.

3. **Productize:** This is the core development phase where the application is built through a series of short, iterative development cycles, typically lasting one to three days each. In each iteration, the researchers implement specific features, conduct unit testing, and integrate the components. During this phase, the dataset is preprocessed, the machine learning models are trained and tested, and the Android application is programmed to perform offline image analysis. Each iteration results in a working, tested increment of the software. If issues are

found, they are corrected within the same or the immediately following iteration, ensuring a consistently functional and improving product.

4. **Stabilize:** After all features are implemented, the researchers focus on integrating and stabilizing all components of the application into a cohesive system. Comprehensive testing is conducted using a separate validation set of banana leaf images to measure system performance using metrics such as Accuracy and Precision. Any bugs, inconsistencies, or integration issues discovered during testing are systematically resolved. This phase ensures that the complete application, including its image processing pipeline and classification models, functions reliably and produces correct diagnostic results before it is released to end-users.
5. **System Test and Fix:** The final phase involves a rigorous and systematic evaluation of the complete application in a real-world context. The finalized Android application is tested by target users, specifically farmers in the field, to gather feedback and identify any remaining issues related to usability or performance under practical conditions. Any critical defects are addressed and corrected. Upon successful completion of this phase, the stable and validated offline Android application is formally released, allowing farmers to use their mobile devices to capture images of banana leaves and receive instant, accurate disease diagnoses and treatment recommendations without requiring internet connectivity.

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